

CHEMISTRY

FORM 3 NOTES



CHEMISTRY FORM THREE REVISION NOTES



ORGANIC CHEMISTRY I

ORGANIC CHEMISTRY

Definition

- The chemistry of hydrogen and carbon chain compounds.
- It the study of carbon compounds except the oxides of carbon i.e. CO, CO₂ and Carbonates.

ORGANIC CHEMISTRY I: THE HYDROCARBONS

Hydrocarbons

Are compounds of hydrogen and carbon only; and are the simplest organic compounds.

Main groups of hydrocarbons

Are classified on the basis of the type of bonds found within the carbon atoms.

- **Alkanes:** Are hydrocarbons in which carbon atoms are linked by single covalent bonds.
- **Alkenes:** Carbon atoms are held by at least one double bond.
- **Alkynes:** Have at least one triple bond between any two carbon atoms.

Saturated and unsaturated hydrocarbons

(a). Saturated hydrocarbons

- Are hydrocarbons which the carbon atoms are bonded to the maximum number of other atoms possible.
- hydrocarbons which don't react and hence cannot decolourise both Bromine water and acidified potassium manganate (VII).
- They are compounds in which each carbon atom has only single covalent bonds, throughout the structure.

(b). Unsaturated hydrocarbons

- Are hydrocarbons which contain at least one double or bond, between any two adjacent carbon atoms.
- The carbon atoms do not have maximum covalency.
- They can decolourise both bromine water and acidified potassium manganate (VII).

Examples: All alkenes and Alkynes.

Experiment: To verify saturated and unsaturated hydrocarbons.

Procedure:

- 3 to 4 drops of bromine water are added to about 1 cm³ of the liquid under investigation.
- The mixture is then shaken thoroughly and the observations recorded;
- For gases the gas under investigation is bubbled into 1 cm³ of bromine water;
- The procedures are then repeated with acidified potassium manganate (VII);

Observations:

COMPOUND	OBSERVATIONS	
	With potassium permanganate	With Bromine water
Kerosene	No observable colour change	No colour change
Laboratory gas	No observable colour change	No observable colour change
Turpentine	Purple colour turns colourless	Solution is decolourised
Hexane	No observable colour change	No observable colour change
Pentene	Potassium permanganate is decolourised	Solution is decolourised



Conclusion

- Kerosene, laboratory gas and hexane are saturated hydrocarbons
- Turpentine and pentane are unsaturated hydrocarbons.

Homologous series

- Refers to a group of organic compounds that have the same general formula, whose consecutive members differ by a similar unit, and usually have similar chemical properties.

Characteristics of a Homologous series.

- Can be represented by a general formula;
 - Have similar chemical properties
 - Have similar structures and names
-
- They show a steady gradation of physical properties
 - Can usually be prepared by similar methods.

Structural and molecular formula

✓ Molecular formulae

- Simply shows the number and type of elements (atoms) in the compound.

✓ Structural formula

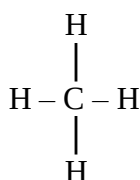
Shows how the different atoms in the molecules (of a compound) are bonded or joined together.

Example:

Methane

Molecular formula CH_4 ;

Structural formula



1. Alkanes

Are the simplest hydrocarbons with the general formula; $\text{C}_n\text{H}_{2n+2}$ where n = number of carbon atoms in the molecule.

Examples:

- For compound with only 1 carbon atom, formula = CH_4
- 2 carbon atoms; the formula = C_2H_6

Names and formulas of the first 10 Alkanes

Note:

Consecutive members of the alkane series differ by a CH_2 -unit, hence a **homologous series**.

(a). General formula

- The Alkanes have a general formula $\text{C}_n\text{H}_{2n+2}$ where n is the number of carbon atoms in the molecule.

**Example:**

When $n = 3$, $(2n + 2) = 8$, and the alkane has the formula C_3H_8 (Propane)

(b). Structure

- In all Alkanes the distribution of bonds around each carbon atom is tetrahedral.

Example: Methane**(c). Homologous series**

- The Alkanes differ from each other by a $-CH_2-$.
- Thus methane, CH_4 differs from ethane, C_2H_6 by $-CH_2-$, and ethane in turn differs from propane C_3H_8 by $-CH_2-$.
- They therefore form a homologous series.

(d). Functional groups

- A functional group is a part of a compound which has a characteristic set of properties.
- Thus when a bromine atom replaces a hydrogen atom in an alkane, it imparts to the compound new chemical and physical properties.

Examples: six important functional groups.

(e). Isomerism

- Is a situation whereby two or more compounds have similar molecular formulae but different structural formula.
- Such compounds are called **isomers**, i.e compounds with the same molecular formula but different structural formula.

Examples: For Butane, (C_4H_{10}) there are **two** possible structures.

Isomers have different physical and chemical properties.

Example: Ethanol and dimethyl ether.

- Molecular formula: both have C_2H_6O

✓ **Structural formula:**

(i). Ethanol

(ii). Dimethyl ether

Differences

Ethanol	Dimethyl ether
- A liquid of boiling point $78.4^\circ C$	- A gas at room temperature (B.P $-24^\circ C$).
- Completely soluble in water	- Slightly soluble in water.
- Reacts with sodium ethoxide and liberates hydrogen gas	- Does not react with sodium metal.

(f). Alkyl groups

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- Is a group formed by the removal of a hydrogen atom from a hydrocarbon.
- Alkyl groups don't exist on their own but are always attached to another atom or group.

Naming of alkyl groups

- Is done by removing the ending -ane from the parent alkane and replacing it with -yl.

Examples

Methane (CH_4) gives rise to Methyl $-\text{CH}_3$

Ethane (C_2H_6) gives rise to ethyl, $-\text{C}_2\text{H}_5$ i.e. $-\text{CH}_2\text{CH}_3$

Propane (C_3H_8) gives rise to Propyl, $-\text{C}_3\text{H}_7$ // $-\text{CH}_2\text{CH}_2\text{CH}_3$;

(g). Nomenclature of Alkanes

- Generally all Alkanes end with the suffix -ane;
- Alkanes can either be **straight chain** or **branched**.

(i). Straight chain Alkanes

- The names of all Alkanes end with the suffix -ane;

Examples:

Methane, ethane, propane, butane.

- With the exception of the first 4 members of the series (i.e. the 4 listed above) the names of Alkanes begin with a Greek prefix indicating the number of carbon atoms in the main chain.

Examples: - Pentane – 5 carbon atoms

Hexane – 6 carbon atoms.

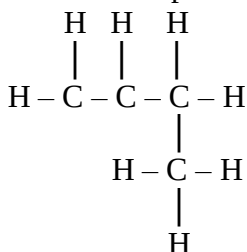
(ii). Branched Alkanes

The naming of branched chain Alkanes is based on the following rules:-

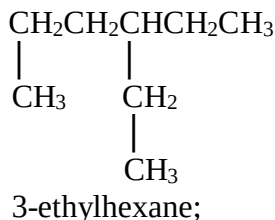
1. The largest continuous chain of carbon atoms in the molecule is used to deduce the parent name of the compound.
2. The carbon atoms of this chain are numbered such that the branching // substituents are attached to the carbon atom bearing the lowest number.
3. The substituent // branch is named e.g. methyl, ethyl etc and the name of the compound written as **one word**.

Examples

Further examples



2-methylpropane;



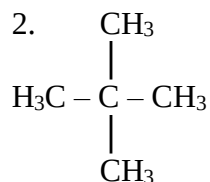
Further examples.

1. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$



3-methylpentane;

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2, 2-dimethylpropane;

Note: refer to course books and draw as many examples as possible.

Draw the structural isomers of:

1. Butane.

2. Pentane;

3. Hexane;

(f). Occurrence of Alkanes

- There are 3 known natural sources:

(i). Natural gas: this consists of mainly of methane;

(ii). Crude oil:

- Consists of a mixture of many Alkanes

- It can be separated into its components by fractional distillation.

Reason:

- The different components have different boiling points.

(iii). Biogas: This contains about 60-75% of methane gas/marshy gas.

Separation of the components of crude oil.

(i). Apparatus

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(ii). Procedure

- The apparatus is arranged as shown above.
- The first distillate appears at about 120°C and is collected, the of 40°C intervals thereafter until the temperatures reach 350°C.

(iii). Observations and explanations

- This method of separation is called fractional distillation, and depends on the fact that the various components of the mixture have **different boiling points**.
- The various fractions vary in properties as explained below.

(a). Appearance

- Intensity of the colour increases with increase in boiling point.
- Boiling point increases with increasing number of carbon atoms.

Reason:

- The higher the number of carbon atoms, the higher the number of **covalent bonds**.
- Thus the first fraction to be distilled (lab gas) is colourless while the last distillates (between) is dark black in colour.

(b). Viscosity

- **Increases** with increasing **boiling point**;
- The fractions with low boiling points are less viscous while the fraction with the highest boiling point is semi-solid;

(c). Inflammability:

- Decreases with increasing boiling points.
- The gaseous fractions, with least boiling points readily catches fire // burn, while the semi-solid fractions with very high boiling points are almost non-combustible.

Note: Some Hydrocarbons are found in more than one fraction of crude oil and more advanced chemical methods are necessary for complete separation.

Uses of the various fractions of crude oil.

No. of carbon atom per molecule	Fractions	Uses
1-4	Gases	Laboratory gases and gas cookers
5-12	Petrol	Fuel in petrol engines
9-16	Kerosene (paraffin)	Fuel for jet engines (aeroplanes) and domestic uses
15-18	Light diesel oils	Fuel for heavy diesel engines e.g. for ships
18-25	Diesel oils	Fuel for diesel engines
20-70	Lubricating oils	Used for smooth running of engine parts



>70	Bitumen	Road tarmacking
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Changes // gradation of physical properties across the alkane homologous series

Name of alkane	Formula	State of room temperature (208K)	M.P (K)	B.P (K)	Density (g cm ⁻³)	Solubility	Solubility
Methane	CH ₄	↑	90	112	0.424		
Ethane	C ₂ H ₆		91	184	0.546		
Propane	C ₃ H ₈	Gaseous	85	231	0.582		
Butane	C ₄ H ₁₀	↓	138	273	0.579		
Pentane	C ₅ H ₁₂	↑	143	309	0.626		
Hexane	C ₆ H ₁₄		178	342	0.659		
Heptane	C ₇ H ₁₆	Liquid		447	0.730		
Octane;	C ₈ H ₁₀						
Nonane	C ₉ H ₂₀						
Decane	C ₁₀ H ₂₂	↓	243				

Preparation and chemical properties of Alkanes

Note:

- Alkanes, like any other Homologous series have similar chemical properties.
- Generally any alkane can be represented form the reaction represented by the following equation:



Thus;

- Methane can be prepared form sodium ethanoate (CH₃COONa)
- Ethane can be prepared form sodium propanoate (CH₃CH₂COONa)
- Propane can be prepared form sodium Butanoate (CH₃CH₂CH₂COONa)

Laboratory Preparation of methane

(i). Apparatus

(ii). Procedure

- About 5g of odium ethanoate and an equal mass of soda lime is put in a hard glass test tube, upon mixing them thoroughly in a mortar.
- The mixture is heated thoroughly in the test-tube.

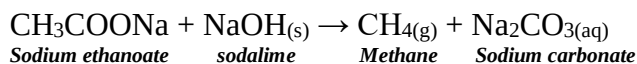
(iii). Observation

- A colourless gas collects over water

Reasons:

- Methane does not react with and is insoluble in water.

Equation



Physical properties of methane

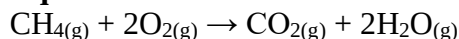
1. It is a non-poisonous, colourless gas.
2. It is slightly soluble in water, but quite soluble in organic solvents such as ethanol and ether.
3. It is less denser than air and when cooled under pressure, it liquefies.

Chemical properties

4. Burning

- It is flammable and burns in excess air // oxygen with a pale blue non-luminous flame to give carbon (IV) oxide and water vapour.

Equation:



Note: In a limited supply of air, the flame is **luminous**.

Reason:

- This is due to incomplete combustion of the methane.
- A mixture of methane and air **explodes** violently when ignited if the volume ratio is approximately 1:10 and this is often the cause of fatal explosions in coal mines.

5. Reaction with Bromine water and acidified potassium permanganate

- When methane is bubbled through bromine water the red brown colour of bromine persists; and when bubbled through acidified potassium manganate (VII) solution; the purple colour of the solution remains;
- Thus it has no effect on either bromine water or acidified potassium permanganate.

Reason: It is a saturated hydrocarbon.

6. Substitution reactions

- A substitution reaction is one in which one atom replaces another atom in a molecule.

Example: The substitution of Bromine in methane.

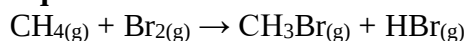
Procedure:

- A sample of Methane (CH_4) is placed in a boiling tube and to it is added some bromine gas.
- The tube is stoppered, and the mixture shaken, then allowed to stand and exposed to **ultra-violet lamp**.

Observations

- The red colour of Bromine begins to fade, and the pungent smell of hydrogen bromide (HBr) gas is detectable when the stopper is removed.
- A moist blue litmus paper also turns **red** on dipping into the resultant mixture.

Equation



Explanation

- For a chemical reaction to occur, bonds must be broken.
- The light energy (V.V. light) splits the Bromine molecule into free atoms, which are very reactive species.
- Similarly the energy breaks the weaker carbon – hydrogen bonds, and not the stronger carbon – carbon bonds.

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- The free bromine atoms can then substitute (replace one of the hydrogen atoms of methane, resulting into bromomethane and hydrogen bromide gas.

Note: This process can be repeated until all hydrogen atoms in CH_4 are replaced.

Write all the equations to show the stepwise substitution of all hydrogen atoms in methane.

- The substitution reactions can also occur with chlorine, forming chloromethane dichloromethane, trichloromethane (chloroform) and tetrachloromethane (carbon tetrachloride) respectively.

Equations:

Uses of methane

- It is used as a fuel
- Used in the manufacture of carbon black which is used in printers ink and paints.
- Used in the manufacture of methanol, methanal, chloromethane and ammonia.

Cracking of Alkanes

- Is the breaking of large alkane molecules into smaller Alkanes, alkenes and often hydrogen.
- It occurs under elevated temperatures of about $400-700^\circ\text{C}$

Equation

Example: Cracking of propane

2. Alkenes

- Are hydrocarbons with at least one carbon-carbon double bond, and have the general formula C_nH_{2n} .
- They thus form a homologous series – with the simplest member behind ethane.

Names and formulae of the first six alkenes.

Name of alkene	Formula
Ethene	
Propane	
Pbut-1-ene	
Pent-lene	
Hex-tene	



NOMENCLATURE OF ALKENES

Rules

1. The parent molecule is the longest carbon chain; and its prefix is followed by the suffix –ene.
2. The carbon atoms in the chain are numbered such that the carbon atoms joined by the double bonds get the lowest possible numbers.
3. The position of the substituent groups is indicated by showing the position of the carbon atom to which they are attached.
4. In case of 2 double bonds in an alkene molecule, the carbon atom to which each double bond is attached must be identified.

Examples

Questions: For each of the following alkenes, draw the structural formula

- i. Hex-1-ene
- ii. Prop-1-ene
- iii. Hex-2-ene

1. Give the IUPAC names for:

Note: Branched alkenes:

Even for branched alkenes, the numbering of the longest carbon chain is done such that the carbon atoms joined by the double bonds get the smallest numbers possible.

Isomerism in alkenes

- Alkenes show two types of isomerism:-

- i. Branching isomerism
 - ii. Positional isomerism
- i) Branching isomerism
Occurs when a substituent group is attached to one of the carbon atoms in the largest chain containing the double bond.

Positional isomerism; in alkenes

Is a situation whereby two or more unsaturated alkenes have same molecular formula but different structural formula; due to alteration of the position of the double bond.

Question: Draw all the possible isomers of Hexene, resulting from positional and branching isomerism.

Gradation of physical properties of Alkenes

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Name of alkene	Formula	(MP ⁰ C)	B.P (0)	Density g/cm ³	solubility
Ethene		-169	-104	-	
Propene		-189	--47.7	-	
But-1-ene		-185	-6.2	-	
Pent-1-ene		-138	-3.0	0.640	
Hex-1-ene		-98	-98	0.674	

Note: the double bond is the reactive site in alkenes

Preparation and chemical properties of Ethene

i) *Apparatus*

ii) *Procedure*

A mixture of ethanol and concentrated sulfuric acid in the ratio 1:2 respectively are heated in a flask to a temp. of 160°C – 180°C.

iii) *Observation*

A colourless gas results; and is collected over water.

Reasons: Its insoluble, unreactive and lighter than water.

iv) *Equation*

v) *Explanation*

At 160°C – 180°C the conc. H₂SO₄ dehydrates the ethanol, removing a water molecule from it and the remaining C and H atoms rearrange and combine to form Ethene which is collected as colourless gas.

Note: At temperature below 140°C, a different compound called ether is predominantly formed.

Ethene can also be prepared by passing hot aluminum oxide over ethanol. The later of which acts as a catalyst i.e.

Reactions of ethene/chemical properties

a) *Burning/combustion*

Just like an alkenes and alkanes, ethene burn in air, producing carbon dioxide and large quantities of heat.

Equation:

Caution: Mixtures of air and ethene can be explosive and must be handled very carefully.

b) *Additional reactions:*

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Is a reaction in which are molecule adds to another to form a single product occur in alkenes due to presence of a double bond.

1. With oxidizing agents

i) Reaction with acidified potassium permanganate.

Procedure: Ethene is bubbled into a test tube containing acidified potassium permanganate.

Observation: The purple colour of the solution disappears.

Explanation: Ethene reduces the potassium permanganate.

The permanganate ion is reduced to Manganese (II) ion and water.

Equation

Note: The net effect of the above reaction is the addition of two –OH groups to the double bond forming ethan-1, 2-dio(ethylene glycol).

In cold countries ethylene glycol is used as an antifreeze in car radiators.

i) Reaction with acidified potassium chromate (VI) ($K_2Cr_2O_7$)

2. Halogenations is the addition of halogen atoms across a double bond.

i) Reaction with Bromine $Br_2(g)$

Procedure: Ethene is mixed with Bromine liquid/gas

Observation: The reddish brown bromine gas is decoloured/becomes colourless.

Explanation: Bromine is decoloured due to the addition of Bromine atoms to the two carbon atoms of the double bond forming 1,2 dibromethane.

ii) Reaction with chlorine

The Chlorine (greenish yellow) also gets decolourised due to the addition of its atoms on the double bond.

Note: Alkenes react with and decolourise halogens and potassium permanganate by additional reaction at room temperature and pressure.

The reaction site is the double bond and hence/all alkenes will react in a similar manner.

Example; Butene and Bromine

iii) Reaction with Bromine water

Bromine is dissolved in water and reacted with ethene.

Equation:

Further examples of additional reactions

3. Addition of hydrogen halides

i) With hydrobromic acid; $HBr(aq)$



With sulphuric acid

4. Addition of Ethene with sulphuric acid

Note: When ethylhydrogen sulphate is hydrolysed, ethanol is formed.

In this reaction, water is added to ethylhydrogen sulphate and the mixture warmed.

5. Ethene with Hydrogen i.e. Hydrogenation.

Is commonly termed hydrogenation though just a typical addition reaction.

Ethene is reacted with hydrogen, under special conditions.

Conditions: moderate temperature and pressure.
Nickel catalyst/palladium catalyst.

Equation:

Application: it is used industrially in the conversion of various oils into fats e.g. in the preparation of Margarine.

6. **Polymerization reactions.**

Also called self-addition reactions

Alkenes have the ability to link together (polymerise) to break the double bond to give a molecule of larger molecular mass (polymers)

Polymers: Are very large molecules formed when 2 or more (smaller) molecules link together to form a larger unit.

Polymers have properties different from those of the original constituent monomers.

Examples: Polymerisation of ethene

i) **Conditions**

- High temperatures of about 200°C
- High/elevated pressures of approximately 1000 atmospheres
- A trace of oxygen catalyst.

ii) **Procedure:** Ethene is heated at 200°C and 1000 atm. Pressure over a catalyst.

iii) **Observation:** Sticky white substance which hardens on cooling is formed.
This solid is called polythene, commonly referred to as polythene.

ii) **Equation:**

Generally

Uses of polythene



1. Used for the manufacture of many domestic articles (bowls, buckets, water cans, and cold water pipes) e.t.c.

Note: Polythene pipes have a great advantage over metal pipes as they can be welded quickly and do not burst in frosty weather.

2. Manufacture of reagent bottles, droppers, stoppers etc. since polythene is unaffected by alkalis and acids.

Test for Alkenes

- They decolourise bromine water, acidified potassium manganate VII.
i.e. These addition reactions show the presence of a double bond.

Uses of Alkenes

1. Manufacture of plastics, through polymerization.
2. Manufacture of ethanol; through hydrolysis reactions
3. Ripening of fruits.
4. Manufacture of ethan – 1, 2-diol(glyco) which is used as a coolant.

3. ALKYNES

Are unsaturated hydrocarbons which form a homologous series of a general formula C_nH_{2n-2} , where $n = 2$ or more.

The functional groups of the alkyne series is the carbon – carbon tripple bond.

They also undergo addition reactions because of High unsaturation and may be polymerised like the alkenes.

Examples

Name	Molecular formula	Structural formular
Ethyne	C_2H_2	CH CH
Propyne	C_3H_4	CH ₃ C CH
But-l-yne	C_4H_6	CH ₃ CH ₂ C CH
Pent-l-yne	C_5H_8	CH ₃ (CH ₂) ₂ C CH

Nomenclature of alkynes

- The largest chain with the tripple carbon – carbon bond forms the parent molecule.
- Numbering of the carbon atoms is done such that the carbon atom with the tripple bond acquires the lowest possible number.
- The substituent branch if any is named, and the compound written as a single word.

Examples



2. Draw the structures of the following hydrocarbons
 - i. 2,2 dimethyl-but-2-yne
 - ii. propyne
 - iii. 4,4 diethyl-hex-2-yne.

Isomerism in alkynes

a) Positional isomerism

Isomerism commonly occurs in alkynes due to the fact that the position of the tripple bond can be altered.

Such isomers, as usual have same molecular but different structural formulas.

Examples

i) Isomers of Butyne

a)

b) Branching isomerism – occurs when alkyl group is present in the molecule.

ii) Others

Gradation in physical properties of Alkynes

Name of Alkyne	Formula	M.P/ ⁰ C	B.P/ ⁰ C	Density/gcm-3
Ethyne	HC CH	-8108	-83.6	-
Propyne	CH ₃ CH	-103	-23.2	-
Butyne	CH ₃ CH ₂ CC CH	-122	8.1	-
Pent-l-yne	CH ₃ CH ₂ CH ₂ C CH	-90	39.3	0.695
Hex-l-yne	CH ₃ (CH ₂) ₃ C CH	-132	71	0.716

Preparation and chemical properties of Ethyne.

a) Preparation

i) Apparatus

ii) Procedure:

Water is dripped over calcium carbide and is collected over water.

Reasons for over-water collection:-

- It's insoluble in water
- Unreactive and lighter than water.

iii) Conditions

- Room temperature

iv) Equation



b) Properties of Ethyne

i) Physical

- Colourless gas, with a sweet smell when pure.
- Insoluble in water and can thus be collected over water.
- Solubility is higher in non-solvents * Draw table on physical properties.

ii) Chemical properties

- Combustion
Ethyne burns with a luminous and very sooty flame; due to the high percentage of carbon content, some of which remains unburnt.
- In excess air, the products are carbon dioxide and water.

Equation

In limited air, they undergoes incomplete combustion, forming a mixture of carbon and carbon dioxide.

Note: A sooty flame observed when a hydrocarbon burns in air is an indication of unsaturation in the hydrocarbon.

Addition reactions

During addition reactions of alkynes (Ethyne) the tripple bond breaks in stages;

a) Reaction with hydrogen (Hydrogenation)

Note: This reaction occurs under special conditions i.e. - Presence of a Nickel catalyst
Temperatures about 200°C

b) Reaction with halogens

i) Reaction with chlorine

ii) With Bromine gas

- The red-brown bromine vapour is decoloured.

Equations

Note: In this reaction Cl_2 should be diluted with an inert.

Reason: Pure Cl_2 reacts explosively with Ethyne, forming carbon and HCl .

c) Reaction with Bromine liquid

When Ethyne reacts with Bromine water, the reddish – brown colour of bromine water disappears.

Reason: The Bromine adds to the carbon tripple bond leading to the of 1;1,2,2 tetrabromoethane.

Equation



E; Ethyne also decolorizes acidified potassium permanganate.

Note: Decolourization of acidified potassium permanganate and bromine water are tests for unsaturated hydrocarbons (alkenes and alkynes)

d) Reaction with hydrogen halides

Uses of Ethyne

1. Industrial manufacture of compounds like adhesives and plastics
2. It's used in the oxy-acetylene flame which is used for welding and cutting metals.

UNIT 2: NITROGEN AND ITS COMPOUNDS.

Unit checklist.

1. Introduction
2. Preparation of nitrogen
 - Isolation from air
 - Isolation from liquid air
 - Laboratory preparation
 - Preparation from ammonia
 - Properties of nitrogen
 - Oxides of nitrogen
 - Nitrogen (I) oxide
 - Nitrogen (II) oxide
 - Nitrogen (IV) oxide
3. Action of heat on nitrates.
4. Ammonia gas
 - Preparation
 - Laboratory preparation
 - Preparation from caustic soda
 - Test for ammonia
 - Fountain experiment
 - Properties and reactions of ammonia
 - Large scale manufacture of ammonia gas: the Haber process
 - Uses of ammonia
5. Nitric (V) acid
 - Laboratory preparation
 - Industrial manufacture of nitric (V) acid: The Ostwald's process.
 - Reactions of dilute nitric acid
 - Reactions of concentrated nitric acid
 - Uses of nitric acid
6. Test for nitrates.
7. Pollution effects of nitrogen and its compounds
8. Reducing pollution environmental pollution by nitrogen compounds.


Introduction:

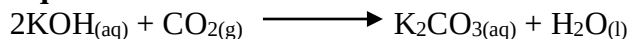
- About 78% of air is nitrogen, existing as N₂ molecules.
- The two atoms in the molecules are firmly held together.
- Nitrogen does not take part in many chemical reactions due to its low reactivity.
- Its presence in air dilutes oxygen and slows down respiration, burning and rusting.

Preparation of nitrogen.
(a). Isolation from air.
(i). Apparatus.
(ii). Procedure.

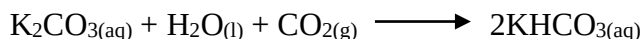
- Air is driven out of the aspirator by passing water into the aspirator from a tap.
- The air is then passed through a wash bottle containing concentrated potassium hydroxide solution.

Reason:

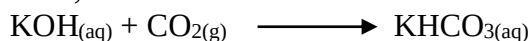
- To remove carbon (IV) oxide from air.

Equations:


Then



Thus;



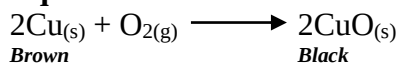
- The carbon (IV) oxide-free air is then passed into a combustion tube with heated copper metal.

Reason:

- To remove oxygen from the air.

Note:

In this reaction the brown copper metal is oxidized to black copper (II) oxide.

Equation:

Note:

- Alternatively oxygen can be removed by passing the carbon (IV) oxide-free air through pyrogallous acid.
- The remaining part of air is mainly nitrogen and is collected over water.

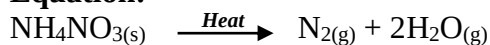
Note:

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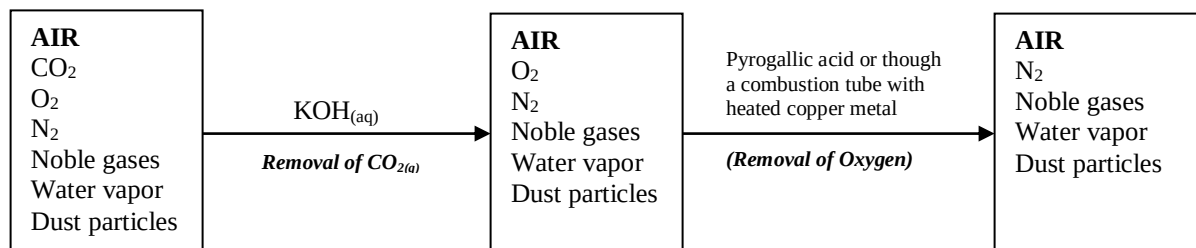


- Nitrogen obtained by this method contains noble gases like xenon, argon etc as impurities.
- Purer nitrogen may be obtained by heating ammonium nitrite.

Equation:



Summary.



(b). Removal from liquid air.

- Liquid air is primarily a mixture of nitrogen and oxygen with small amounts of noble gases.
- This method involves manufacture of liquid air and consequent fractional distillation.

The chemical process.

Step 1: removal of dust particles.

- Dust particles are first removed by either of the two processes:
 - Electrostatic precipitation
 - Filtration.

(i). Electrostatic precipitation:

- Air is passed through oppositely charged plates hence an electric field.
- Dust particles (charged) are consequently attracted to plates of opposite charges.

Diagram: electrostatic precipitation:

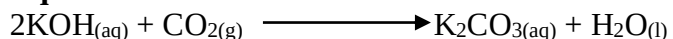
(ii). Filtration:

- The air is passed through a series of filters which traps dust particles as the air is forced through.

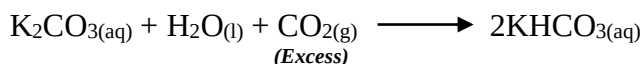
Step 2: removal of carbon (IV) oxide.

- The dust-free air is passed through a solution of potassium hydroxide; to remove carbon (IV) oxide.

Equations:



Then:



- Alternatively, sodium hydroxide may be used in place of potassium hydroxide.

Step 3: Removal of water vapour.

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- The dustless, carbon (IV) oxide-free air is next passed into a chamber with **concentrated sulphuric acid** or **anhydrous calcium chloride** in which water vapour is separated and removed.

Note:

To remove water vapour, air may be alternatively passed into a **freezing chamber** where it is condensed at -25°C .

- The water vapour **solidifies** and is then absorbed by **silica gel** and separated out.
- Air is freed from carbon (IV) oxide, water vapour and dust particles (before compression) so as to prevent **blockage** of the pipes caused by solid materials at liquefaction temperatures i.e. carbon (IV) oxide and water vapour form solids which may block the collection pipes.

Step 4: Liquefaction of air.

- The air free from dust, carbon (IV) oxide and water vapour is then compressed at about 200 atmospheres, cooled and allowed to expand through fine jet.
- This sudden expansion causes further cooling and the gases eventually liquefy.
- The liquid is tapped off through a valve while gas which has escaped liquefaction returns to the compressor.
- Liquid air is a transparent **pale blue liquid**.
- This liquid is then fractionally distilled.

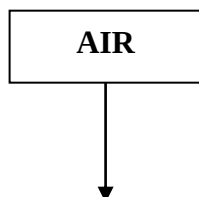
Step 5: Fractional distillation of liquid air.

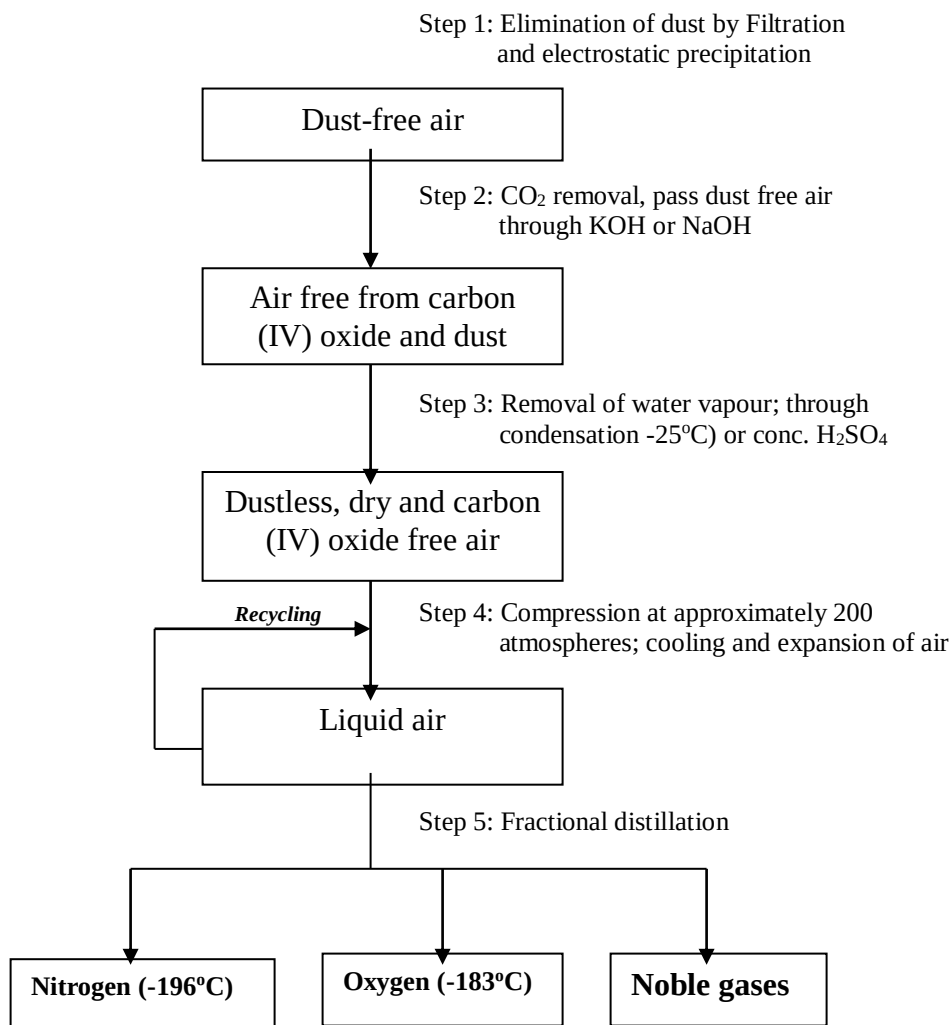
- The boiling point of nitrogen is -196°C (77K) and that of oxygen is -183°C (90K).
- Consequently when liquid air is allowed to warm up, the nitrogen boils off first and the remaining liquid becomes richer in oxygen.
- The top of the fractionating column is a few degrees cooler than the bottom.
- Oxygen, the liquid with the higher boiling point (-183°C) collects at the bottom as the liquid.
- The gas at the top of the column is nitrogen which has a lower boiling point.
- The more easily vapourised nitrogen is taken off.
- This way about 99.57% nitrogen is obtained.

Note:

- The separation of nitrogen and oxygen from air is a proof that air is a mixture and not a compound.

Summary: Fractional distillation of liquid air.





(c). Laboratory preparation method.

(i). Apparatus.

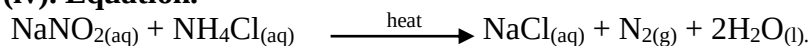
(ii). Procedure:

- Concentrated solutions of sodium nitrite and ammonium chloride are heated together in a round bottomed flask.

(iii). Observations.

- Colourless gas (nitrogen) is evolved rapidly and is collected over water.

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**(iv). Equation.**

Note: the resultant gas is less dense than that isolated from air.

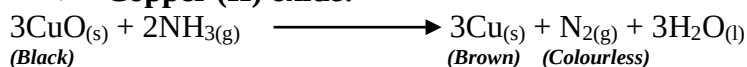
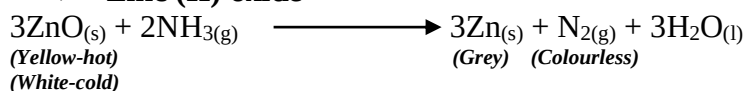
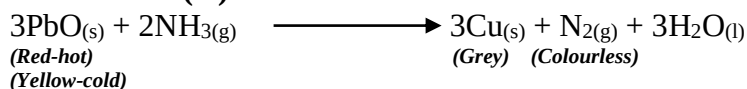
Reason:

- It does not contain impurities.

(d). Preparation from ammonia.**(i). Apparatus.****(ii). Procedure:**

- Dry ammonia gas is passed over a heated metal oxide e.g. copper metal.
- The **metal oxide** is **reduced** to the metal while **ammonia gas** is itself **oxidized** to **nitrogen** and **water**.
- Water is condensed and collected in a u-tube immersed in ice cubes.
- Nitrogen produced is collected **over water**.

(iii). Observations and explanations.

➤ **Copper (II) oxide:**➤ **Zinc (II) oxide**➤ **Lead (II) oxide****Properties of nitrogen.****(a). Physical properties.**

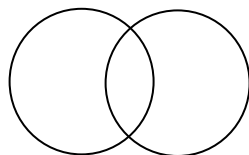
1. It is a colourless, odourless and tasteless gas; almost completely insoluble in water.
2. Slightly lighter than air.

(b). Chemical properties.

3. It is inert (unreactive)

**Reason:**

- The inert nature of nitrogen is due to the strong covalent bonds between the two nitrogen atoms in the molecule; N₂.

Structurally;

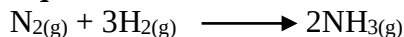
- In air, it neither burns nor supports combustion and acts mainly as a diluent for the oxygen; slowing down the rate of burning.

Chemical test for nitrogen.

- A gas is proved to be nitrogen by elimination: -
 - It extinguishes a lighted splint and does not burn; hence it is not oxygen, hydrogen or carbon (II) oxide.
 - It has neither **smell** nor **colour**; and therefore is not chlorine, ammonia, sulphur (IV) oxide or hydrogen chloride.
 - It does not form a **white precipitate** in lime water, and so it is not carbon (IV) oxide.
 - It is **neutral** to litmus and therefore cannot be carbon (IV) oxide, hydrogen sulphide, ammonia, hydrogen chloride

4. Reaction with hydrogen.

- Under special conditions (i.e. high pressure, low temperatures and presence of iron catalyst), nitrogen combines with hydrogen to produce ammonia.

Equation:

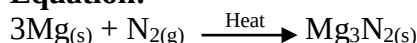
- This reaction forms the basis of **Haber process** used in the manufacture of ammonia.

5. Reaction with burning magnesium.**(i). Apparatus.****(ii). Procedure:**

- A piece of burning magnesium ribbon is introduced into a gas jar full of nitrogen.

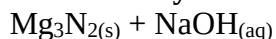
(iii). Observations:

- The magnesium ribbon continues to burn and a **white solid**; magnesium nitride is formed.

Equation:

**Note:**

- When magnesium nitride is treated with water or a solution of sodium hydroxide; the characteristic pungent smell of ammonia can be detected.

Equations:**In water****In sodium hydroxide:****6. Reaction with oxygen.**

- When nitrogen and oxygen in air are passed through an electric arc small quantities of nitrogen (II) oxide are formed.

Equation:**Note:**

- Nitrogen reacts with oxygen under various conditions to give different types of nitrogen oxides.

Uses of nitrogen

1. Used in the **Haber process** in the manufacture of **ammonia**.
2. Due to its **inert nature**, it is mixed with argon to fill electric bulbs (to avoid soot formation).
3. In liquid state it is used as an inert **refrigerant** e.g. storage of semen for artificial insemination.
4. Due to its **inert nature**, it is used in food preservation particularly for canned products i.e. it prevents combination of oxygen and oil which tends to enhance rusting.
5. It is used in oil field operation called enhanced oil recovery where it helps to force oil from subterranean deposits.

Oxides of nitrogen.

- The three main oxides of nitrogen are:

- Nitrogen (I) oxide, N_2O
- Nitrogen (II) oxide, NO
- Nitrogen (IV) oxide, NO_2

1. Nitrogen (I) oxide.**Preparation of nitrogen (I) oxide, N_2O** **(i). Apparatus.**

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**(ii). Procedure:**

- Ammonium nitrate is gently heated in a boiling tube and gas produced collected over warm water.
- Heating is stopped while excess ammonium nitrate still remains.

Reason:

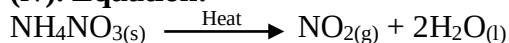
- To avoid chances of an **explosion**.

(iii). Observations:

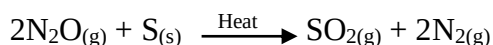
- The solid (ammonium nitrate) **melts** and gives off nitrogen (I) oxide which is collected over **warm water**.

Reasons:

- Nitrogen (I) oxide is **slightly soluble** in cold water.

(iv). Equation:**Properties:**

1. It is a colourless gas, denser than air, fairly soluble in cold water and neutral to litmus.
2. It supports combustion by **oxidizing** elements like sulphur, magnesium and phosphorus under strong heat.

Equations:

3. Magnesium **decomposes** the gas and continues to burn in it.

Equation:

4. When exposed over red-hot finely divided copper it is reduced to **nitrogen**.

Equation:

5. Chemical test.

- It **relights** a glowing splint.

**Note:**

- It can be distinguished from oxygen by the following tests:
- It has a **sweet sickly smell**; oxygen is odourless.
- It will not give **brown fumes** (NO₂) with nitrogen (II) oxide; oxygen does.
- It is **fairly soluble** in cold water; oxygen is insoluble.
- It extinguishes feebly burning **sulphur**; oxygen does not.

Uses of nitrogen (I) oxide.

- It was formerly used in hospitals as an **aesthetic** for **dental surgery** but has since been discontinued due to availability of more efficient anaesthetics.

Note:

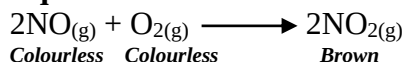
- Nitrogen (I) oxide is also called **laughing gas**; because patients regaining consciousness from its effects may laugh hysterically.

2. Nitrogen (II) oxide, NO.**Preparation:****(i). Apparatus.****(ii). Procedure:**

- Action of heat on 50% concentrated nitric acid on copper turnings.
- Not any heat is required.

Equation:**(iii). Observations:**

- An **effervescence** occurs in the flask; with **brown fumes** because the nitrogen (II) oxide produced reacts with oxygen of the air in the flask to form a brown gas, nitrogen (IV) oxide.

Equation:

- The brown fumes eventually disappear and the gas collected over water.
- The NO₂ fumes dissolve in the water in the trough, resulting into an acidic solution of nitrous acid.
- The residue in the flask is a **green** solution of **copper (II) nitrate**.

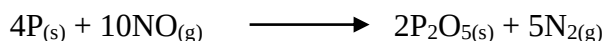
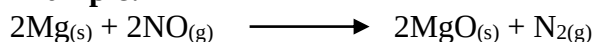


- Industrially, the gas is obtained when ammonia reacts with oxygen in the presence of platinum catalyst.
- This is the first stage in the production of nitric acid.

(v). Properties.

1. It is a colourless, **insoluble** and **neutral** to litmus. It is also slightly denser than air.
2. Readily combines with oxygen in air and forms brown fumes of nitrogen (IV) oxide.
3. Does not support combustion except in the case of **strongly burning magnesium** and **phosphorus**; which continues to burn in it, thereby reducing it i.e. it is an oxidizing agent.

Example:



4. When passed over red-hot finely divided copper, it is reduced to nitrogen gas.

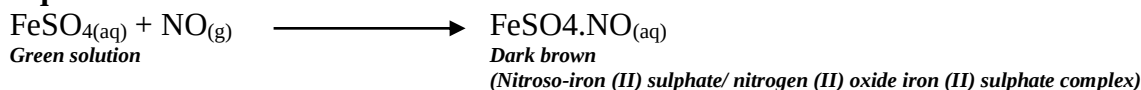
Equation:



5. Reaction with iron (II) sulphate.

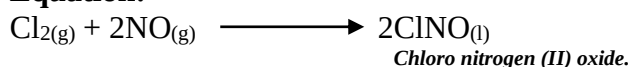
- When iron (II) sulphate solution (freshly prepared) is poured into a gas jar of nitrogen (II) oxide, a dark brown colouration of **Nitroso-iron (II) sulphate is obtained**.

Equation:



6. It is also a reducing agent.

Equation:



7. Reaction with hydrogen.

- When electrically sparked with hydrogen, NO is reduced to **nitrogen**.

Equation:



Chemical test:

- When exposed to air, nitrogen (II) oxide forms brown fumes of nitrogen (IV) oxide.

Uses of Nitrogen (II) oxide.

Note: -It is not easy to handle owing to its ease of **oxidation**.

1. It is an intermediate material in the manufacture of nitric acid

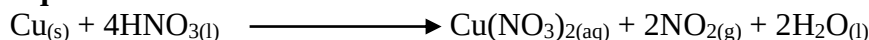
3. Nitrogen (IV) oxide.

Preparation:

(i). Apparatus.

**(ii). Procedure:**

- Action of conc. Nitric acid on copper metal.

Equation:**Note:**

- NO₂ is also prepared by the action of heat on **nitrates of heavy metals** like lead nitrate.
- NO₂ is given off together with **oxygen** when nitrates of heavy metals are heated.
- It is best prepared by heating **lead (II) nitrate** in a hard glass test tube.
 - Lead (II) nitrate is the most suitable because it crystallizes without **water of crystallization** (like other nitrates) which would interfere with preparation of nitrogen (IV) oxide that is **soluble** in water.
- The gas evolved passes into a U-tube immersed in an ice-salt mixture.

- **Apparatus:**

- **Equation:**

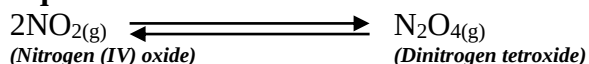


- **Observations:**

- The heated **white** lead (II) nitrate crystals decompose and **decrepitate** (cracking sound) to yield **red** lead (II) oxide; which turns **yellow** on cooling.
- A **colourless gas**, oxygen is liberated, followed immediately by **brown fumes** of nitrogen (IV) oxide.
- Nitrogen (IV) oxide is condensed as a **yellow liquid; dinitrogen tetroxide** (N₂O₄); and is collected in the U-tube.

Note:

- At room temperature, nitrogen (IV) oxide consists of **nitrogen (IV) oxide** and **dinitrogen tetroxide** in equilibrium with each other.

Equation:

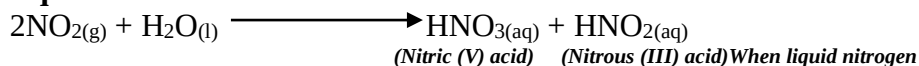


- The oxygen being liberated does not condense because it has a low boiling point of -183°C .

Properties of nitrogen (IV) oxide.

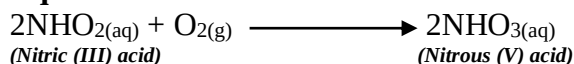
1. **Red-brown** gas with a pungent choking smell
2. It is extremely poisonous.
3. It is **acidic**, hence turns moist litmus paper red.
4. When reacted with water, the brown fumes dissolve showing that it is readily **soluble in water**.

Equation:



- Like carbonic (IV) acid, nitrous (III) acid could not be isolated. It is easily oxidized to nitric (V) acid.

Equation:



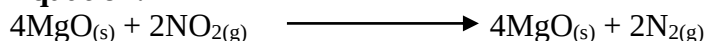
5. Reaction with magnesium.

- Nitrogen (IV) oxide does not support combustion.
- However burning magnesium continues to burn in it.

Reason:

- The high **heat of combustion** of burning magnesium decomposes the nitrogen (IV) oxide to nitrogen and oxygen; the oxygen then supports the burning of the magnesium.

Equation:

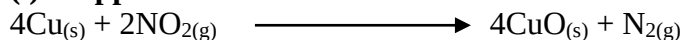


Note:

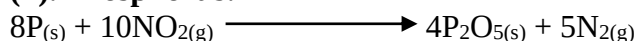
- Generally nitrogen (IV) oxide oxidizes hot metals and non-metals to oxides and itself reduced to nitrogen gas.

Examples:

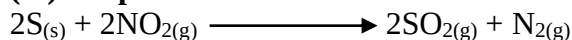
(i). Copper:



(ii). Phosphorus:



(iii). Sulphur:



Note:

- NO_2 reacts with burning substances because the heat decomposes it to NO_2 and O_2 .

Equation:



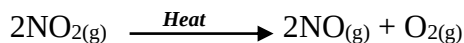
- This is the oxidizing property of nitrogen (IV) oxide.
- The resultant oxygen supports the burning.

6. Effects of heat:

- On heating, nitrogen (IV) oxide dissociates to nitrogen (II) oxide and oxygen and will support a burning splint.

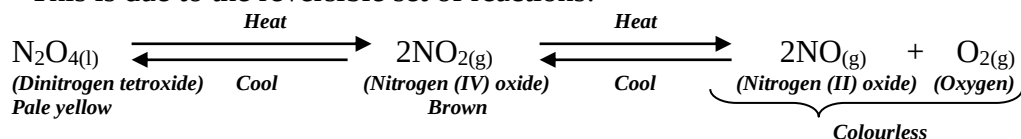
Equation:

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7. - When liquid nitrogen (IV) oxide or dinitrogen tetroxide is warmed, it produces a **pale brown vapour**.

- This is due to the reversible set of reactions:



- Percentage of each in the equilibrium depends on temperature.
- At low temperatures, percentage of N_2O_4 is high and the mixture is pale yellow in colour.
- Percentage of nitrogen (IV) oxide increases with increase in temperature and the colour darkens till at 150°C when the gas is entirely NO_2 and is almost black.
- Still at higher temperatures, nitrogen (IV) oxide dissociates into colourless gas (NO and O_2).

8. Reaction with alkalis.

- A solution of aqueous sodium hydroxide is added to a gas jar of nitrogen (IV) oxide and shaken.

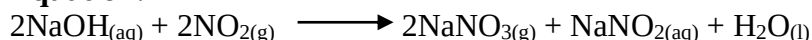
Observation:

- The brown fumes disappear.

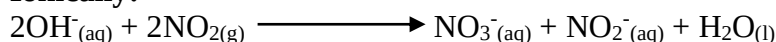
Explanation:

- Formation of sodium nitrate and sodium nitrite.

Equation:



Ionically:



Conclusion:

Nitrogen (IV) oxide is an acidic gas because it can react with an alkali.

Uses of nitrogen (IV) oxide.

1. Mainly used in the manufacture of nitric (V) acid.

Summary on comparison between oxides of nitrogen.

	Nitrogen (I) oxide	Nitrogen (II) oxide	Nitrogen (IV) oxide
Colour	- Colourless gas - Sweet sickly smell	- Colourless; turns brown in air; - Odourless	- Red brown gas; - Choking pungent smell;
2. Solubility	- Fairly soluble in cold water; but less soluble in hot water;	- Almost insoluble in water	- Readily soluble in water to form nitric (V) acid and nitrous (III) acid;
3. Action on litmus	- Neutral to litmus	- Neutral to litmus	- Turns moist blue litmus paper red; i.e. acidic.
4. Combustion	- Supports combustion; relights a glowing splint;	- Does not support combustion;	- Does not support combustion.
5. Density	- Denser than air	- Slightly denser than air	- Denser than air;
6. Raw materials and conditions	- Ammonium nitrate and heat;	- Copper and 50% nitric acid;	- Copper metal and concentrated nitric acid;

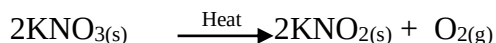
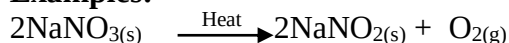


Action of heat on nitrates.

- All nitrates except ammonium nitrate decompose on heating to produce oxygen gas as one of the products.
- Nitrates can be categorized into 4 categories based on the products formed when they are heated.
- The ease with which nitrates decompose increases down the electrochemical series of metals.

1. Nitrates of metals higher in the electrochemical series like sodium and potassium decompose on heating to give the corresponding metal nitrite and oxygen.

Examples:



2. Nitrates of most other metals (heavy metals) that are average in the electrochemical series decompose on heating to give the metal oxide; nitrogen (IV) oxide and oxygen gas.

Example: action of heat on lead (II) nitrate.

(i). Apparatus:

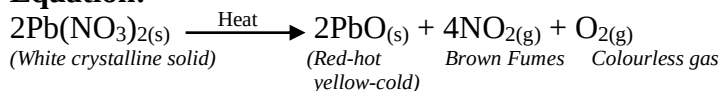
(ii). Procedure:

- Solid white lead (II) nitrate crystals are strongly heated in a boiling (ignition) tube.
- Products are passed into a U- tube immersed in ice.
- Excess gases are channeled out to a fume chamber.

(iii). Observations:

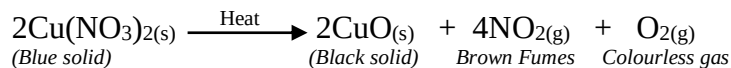
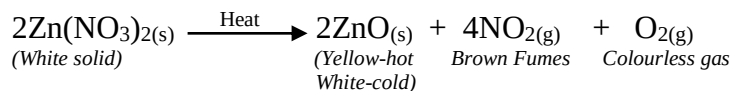
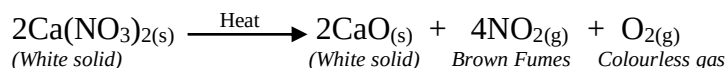
- The white crystalline solid decrepitates.
- A colourless gas (**oxygen**) is liberated and immediately followed by a red brown fumes/ gas (**nitrogen (IV) oxide**).
- A pale yellow liquid (**dinitrogen tetroxide**) condenses in the U-tube in the ice cubes.
- This is due to condensation of nitrogen (IV) oxide.
- A residue which is **red** when hot and **yellow** on cooling remains in the boiling (ignition) tube

Equation:



Further examples:

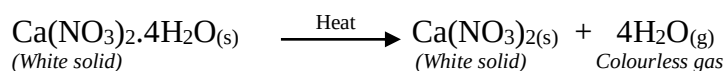
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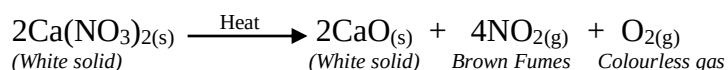
Note:

- Some nitrates are **hydrated** and when heated first give out their water of crystallization; and then proceed to as usual on further heating.

Examples:

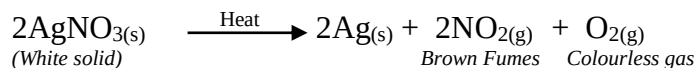
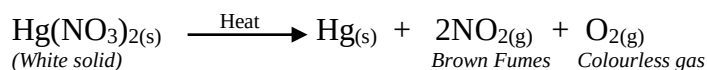


On further heating;



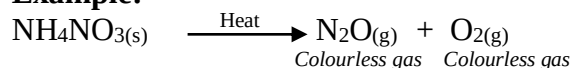
3. Nitrates of metals lower in the reactivity series e.g. **mercury** and **silver** decompose on heating to give the metal, nitrogen (IV) oxide and oxygen.

Example:



4. Ammonium nitrate decomposes to nitrogen (I) oxide and water vapour.

Example:



Note:

This reaction is potentially dangerous as ammonium nitrate **explodes** on strong heating.

Ammonia.

- Is a compound of nitrogen and hydrogen and is the most important hydride of nitrogen.
- It is formed when any **ammonium salt** is heated with an **alkali** whether in solid or solution form.

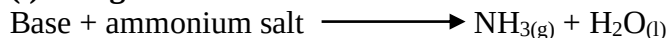
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- It is a **colourless** gas with a pungent smell of **urine**.
- It is **alkaline** and turns moist red litmus paper to blue when introduced to it.

Laboratory preparation of ammonia.

(i). Reagents.



(ii). Apparatus.

(iii). Procedure.

- Ammonium chloride (NH_4Cl)/ sal-ammoniac is mixed with a little dry slaked lime i.e. $\text{Ca}(\text{OH})_2$ and the mixture thoroughly ground in a mortar.

Reason:

- To increase surface area for the reactions.

- The mixture is then heated in a round-bottomed flask.

Note:

- A round-bottomed flask ensures **uniform distribution** of heat while heating the reagents.
- The flask should not be **thin-walled**.

Reason:

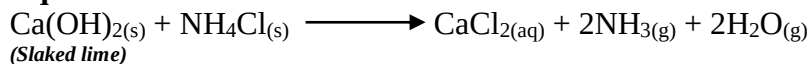
The pressure of ammonia gas liberated during heating may easily **crack or break** it.

- The flask is positioned **slanting downwards**.

Reason:

- So that as water condenses from the reaction, it does not run back to the hot parts of the flask and crack it.
- The mixture on heating produces ammonia, calcium chloride and water.

Equation:



(iv). Drying:

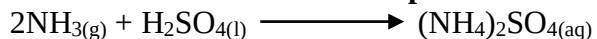
- Ammonia is dried by passing it through a tower or U-tube filled with quicklime (calcium oxide) or pellets of caustic potash but not caustic soda which is **deliquescent**.

Note:

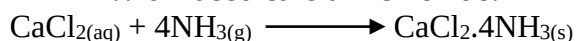


Ammonia cannot be dried with the usual drying agents; concentrated sulphuric acid and calcium chloride as it reacts with them.

➤ **With concentrated sulphuric acid.**



➤ **With fused calcium chloride:**



- i.e. ammonia reacts forming **complex ammonium salt**.

(v). Collection:

- Dry ammonia gas is collected by **upward delivery**.

Reasons:

- It is **lighter** than air.
- It is **soluble** in water.

Other methods of preparing ammonia.

(b). Ammonia from caustic soda (sodium hydroxide) or caustic potash (potassium hydroxide)

Note:

- The slaked lime is replaced by either of the above solutions.
- Thus the solid reactant is ammonium chloride and the liquid reactant is potassium hydroxide.

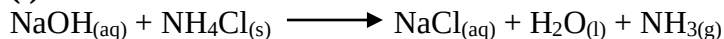
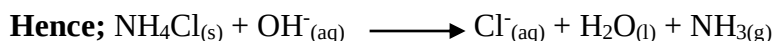
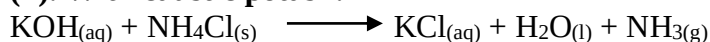
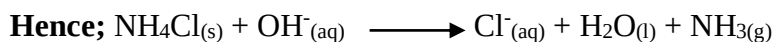
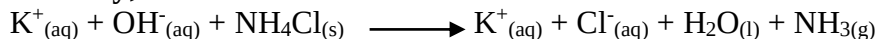
(i). Apparatus:

**(ii). Procedure:**

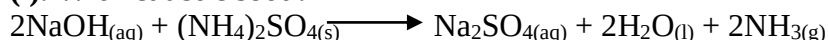
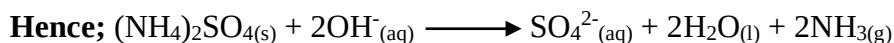
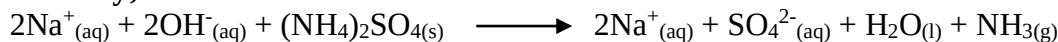
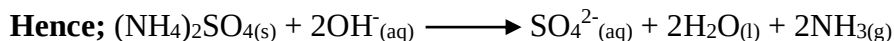
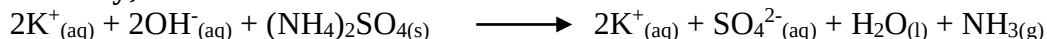
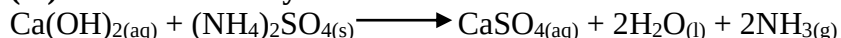
- The flask is not slanted. It is vertical and heated on a tripod stand and wire gauze.

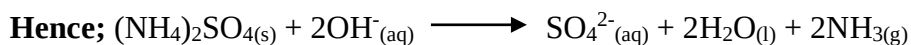
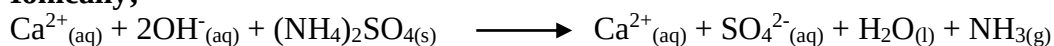
Reason:

- No need of slanting since water produced is in liquid form and not gaseous. Thus there is no possibility of condensation of water on hotter parts.

Equations:**(i). With caustic soda:****Ionically;****(ii). With caustic potash:****Ionically;****Note:**

Ammonium sulphate could be used in place of ammonium chloride in either case.

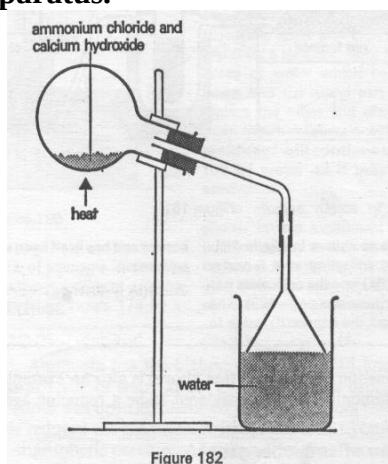
Equations:**(i). With caustic soda:****Ionically;****(ii). With caustic potash:****Ionically;****(iii). With calcium hydroxide:**

Ionicall;**Note:**

Reaction with calcium hydroxide however stops prematurely, almost as soon as the reaction starts.

Reason;

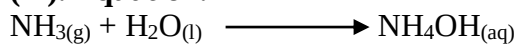
- Formation of insoluble calcium sulphate which coats the ammonium sulphate preventing further reaction.

Preparation of ammonium solution.**(i). Apparatus.****(ii). Procedure:**

- The apparatus is altered as above.
- The drying tower is removed and the gas produced is directly passed into water by an inverted funnel.

Reasons for the inverted broad funnel.

- It increases the **surface area** for the **dissolution** of thereby preventing water from “**sucking back**” into the hot flask and hence prevents chances of an **explosion**.

(iii). Equation.**Note:**

- The solution cannot be prepared by leading the gas directly to water by the delivery tube.

Reason:

- Ammonia gas is **very soluble** in water and so water would rush up the delivery tube and into the hot flask causing it to crack.
- The rim of the inverted funnel is **just below** the water surface.

Tests for ammonia.

1. It is a colourless gas with a **pungent smell**.
2. It is the only common gas that is **alkaline** as it turns moist red litmus paper blue.
3. When ammonia is brought into contact with hydrogen chloride gas, **dense white fumes** of ammonium chloride are formed.

**Equation:****Fountain experiment.****(i). Diagram:****(ii). Procedure:**

- Dry ammonia is collected in a round-bottomed flask and set up as above.
- The clip is open and solution let to rise up the tube.
- The clip is closed when the solution reaches the top of the tube after which it is again opened for a while.

(iii). Observations and explanations.

- When a drop of water gets to the jet, it dissolves so much of the ammonia gas that a **partial vacuum** is created inside the flask.
- As the ammonia in the flask dissolves, the pressure in the flask is **greatly reduced**.
- The atmospheric pressure on the water surface in the beaker forces water into the flask vigorously.
- The drawn-out jet of the tube causes a **fountain** to be produced.
- The fountain appears **blue** due to the **alkaline** nature of ammonia.

(iv). Caution:

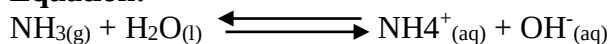
- Ammonia is highly soluble in water forming an alkaline solution of ammonium hydroxide.

Note:

1 volume of water dissolves about 750 volumes of ammonia at room temperature.

Properties and reactions of ammonia.

1. **Smell:** has a characteristic pungent smell.
2. **Solubility:** it is highly soluble in water. The dissolved ammonia molecule reacts partially with water to form ammonium ions (NH_4^+) and hydroxyl ions (OH^-)

Equation:

- Formation of hydroxyl ions means that the aqueous solution of ammonia is (weakly) alkaline and turns universal indicator purple.



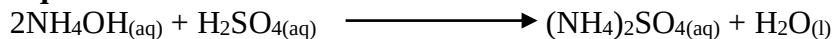
3. Reaction with acids.

- Sulphuric acid and concentrated ammonia solution are put in a dish and heated slowly.
- The mixture is evaporated to dryness.

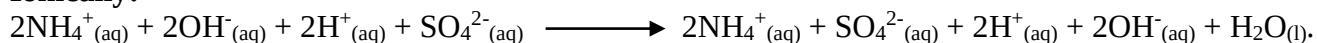
Observations:

- A white solid is formed.

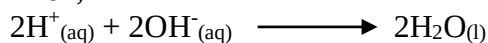
Equation:



Ionic:



Then;

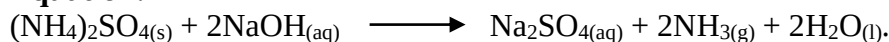


- To some of the resultant white solid, a little $\text{NaOH}_{(\text{aq})}$ was added and the mixture warmed.
- The gas evolved was tested for ammonia.

Observation:

- The resultant gas tested positive for ammonia.

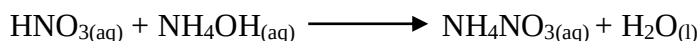
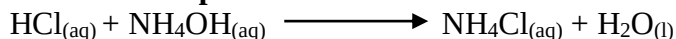
Equation:



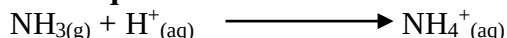
Explanations:

- Evolution of ammonia shows that the white solid formed is an **ammonium salt**.
- The ammonia reacts with acids to form ammonium salt and water only.

Further examples:



Ionic equation:



4. Reaction of ammonia with oxygen.

- Ammonia **extinguishes** a lighted taper because it does not support burning.
- It is **non-combustible**.
- However it burns in air enriched with oxygen with a **green-yellow flame**.

Experiment: Burning ammonia in oxygen.

(i). Apparatus.

(ii). Procedure:

- Dry oxygen is passed in the U-tube for a while to drive out air.
- Dry ammonia gas is then passed into the tube.
- A lighted splint is then passed into the tube.

(iii). Observations:

- A colourless gas is liberated.
- Droplets of a colourless liquid collect on cooler parts of the tube.

(iv). Explanations:

- The conditions for the reactions are:
 - Dry ammonia and oxygen gas i.e. the gases must be **dry**.
 - All air must be driven out of the tube.
- Ammonia burns continuously in oxygen (air enriched with oxygen) forming nitrogen and water vapour i.e. ammonia is oxidized as hydrogen is removed from it leaving nitrogen.

Equation:**Sample question:**

Suggest the role of glass wool in the tube.

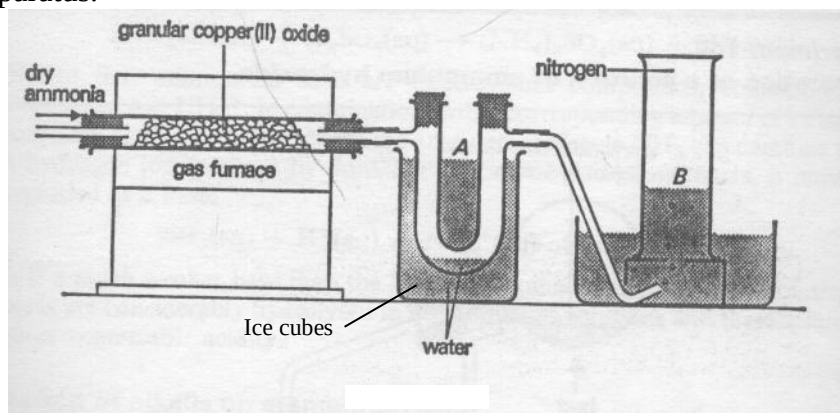
Solution:

- To slow down the escape of oxygen in the combustion tube, thus providing more time for combustion of ammonia.

5. Ammonia as a reducing agent.

- It reduces oxides of metals below iron in the reactivity series.

Experiment: reaction between ammonia and copper (II) oxide.

(i). Apparatus.**(ii). Procedure:**

- Copper (II) oxide is heated strongly and dry ammonia is passed over it.
- The products are then passed through a U-tube immersed in cold water (ice cubes).

(iii). Observations.

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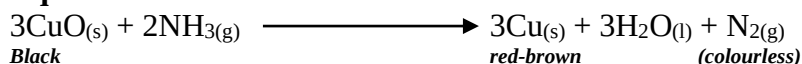


- The copper (II) oxide **glows** as the reaction is **exothermic**.
- A **colourless liquid** collects in the U-tube.
- A **colourless gas** is collected over water.
- The **black** copper (II) oxide changes to **brown** copper metal.

(iv). Explanations.

- Ammonia gas **reduces** copper (II) oxide to copper and is itself **oxidized** to nitrogen and water.

Equation:



- The water produced condenses in the U-tube immersed in cold (ice) water.
- The resultant nitrogen is collected by downward displacement of water.
- The nitrogen gas collected is ascertained indirectly as follows:
 - A lighted splint is extinguished and the gas does not burn; thus it is not oxygen, hydrogen, or carbon (II) oxide.
 - It has neither smell nor colour; it is not ammonia, chlorine, sulphur (IV) oxide or nitrogen (IV) oxide.
 - It is not carbon (II) oxide because it does not turn lime water into a white precipitate.

Note:

- This experiment proves that ammonia contains nitrogen.

6. Reaction with chlorine.

(i). Procedure:

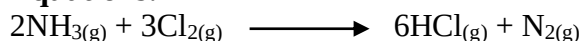
- Ammonia gas is passed into a bell jar containing chlorine.

(ii). Apparatus:

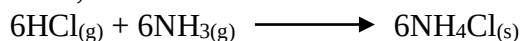
(iii). Observations:

- The ammonia catches fire and burns for a while at the end of the tube.
- The flame then goes out and the jar then gets filled with dense white fumes of ammonium chloride.

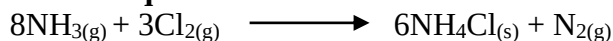
Equations:



Then;



Overall equation:

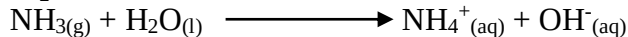




7. Ammonia solution as an alkali.

- Solution of ammonia in water contains hydroxyl ions.

Equation:



- Thus it has many properties of a **typical alkali**.
- Ammonia salts are similar to metallic salts.
- The group (NH_4^+) precipitates in the reaction as a whole without splitting in any way.
- It exhibits unit valency in its compounds and therefore called a **basic radical**.

Note:

- It cannot exist freely as ammonia gas (NH_3) which is a compound.
- Like other alkalis, ammonia solution precipitates insoluble metallic hydroxides by double decomposition when mixed with solution of salts of the metals.

8. Reaction with air in the presence of platinum wire.

(i). Apparatus:

(ii). Procedure:

- Concentrated ammonia solution is put in a conical flask.
- The platinum (or even copper) wire is heated until white-hot.
- Oxygen gas or air is then passed through the ammonia solution.
- The red-hot platinum (copper) wire is then put into the flask containing the concentrated ammonia.

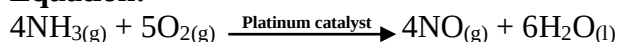
(iii). Observations:

- The hot platinum wire **glows**.
- Red-brown fumes are evolved.

(iv). Explanations:

- The hot platinum coil glows when it comes into contact with the ammonia fumes, which come from the concentrated ammonia solution.
- Reaction between ammonia and oxygen takes place on the **surface of the platinum** wire that acts as a **catalyst**.
- A lot of **heat** is produced in the reaction that enables the platinum coil to continue glowing.
- Ammonia is oxidized to **nitrogen (IV) oxide**.

Equation:

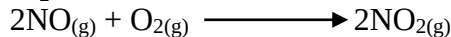


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- Red-brown fumes of nitrogen (IV) oxide are produced due to further **oxidation** of the nitrogen (II) oxide to form nitrogen (IV) oxide.

Equation:



9. Action of aqueous ammonia on solution of metallic salts

(i). Procedure:

- To about 2cm³ of solutions containing ions of calcium, magnesium, aluminium, zinc, iron, lead, copper etc in separate test tubes; aqueous ammonia is added dropwise till in excess.

(ii). Observations:

The various metal ions reacted as summarized in the table below.

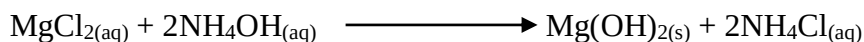
Metal ions in solution	Observations on addition of ammonia	
	Few drops of ammonia	Excess drops of ammonia
Ca ²⁺	White precipitate	White precipitate persists;
Mg ²⁺	White precipitate	Precipitate persists;
Al ³⁺	White precipitate	Precipitate persists;
Zn ²⁺	White precipitate	Precipitate dissolves;
Fe ²⁺	Pale green precipitate	Precipitate persists; slowly turns red-brown on exposure to air;
Fe ³⁺	Red-brown precipitate	Precipitate persists;
Pb ²⁺	White precipitate	Precipitate persists;
Cu ²⁺	Pale blue precipitate	Precipitate dissolves forming a deep blue solution;

(iii). Explanations:

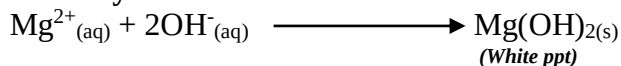
- Most metal ions in solution react with ammonia solution to form insoluble metal hydroxides.
- In excess ammonia, some of the so formed hydroxides dissolve forming complex ions.

(iv). Equations:

1. Mg²⁺_(aq) from MgCl₂;

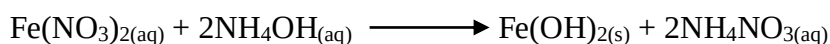


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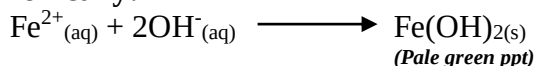




2. Fe^{2+} from $\text{Fe}(\text{NO}_3)_2$;

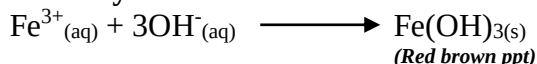


Ionically:



3. Fe^{3+} from FeCl_3 ;

Ionically:

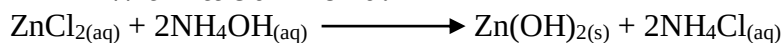


Note:

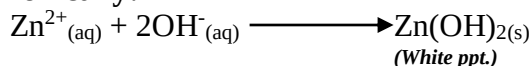
$\text{Zn}^{2+}_{(\text{aq})}$ and $\text{Cu}^{2+}_{(\text{aq})}$ dissolve in excess ammonia solution forming complex ions.

4. Zinc ions and ammonia solution.

➤ **With little ammonia:**



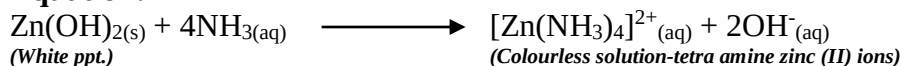
Ionically:



➤ **In excess ammonia:**

- The white precipitate of $\text{Zn}(\text{OH})_{2(\text{s})}$ dissolves in excess ammonia to form a colourless solution; proof that solution has Zn^{2+} ions;
- The colourless solution is a complex salt of tetra-amine zinc (II) ions.

Equation:

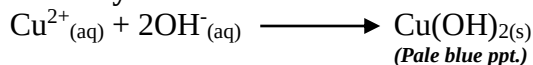


5. Copper (II) ions.

➤ **With little ammonia:**

- A pale blue precipitate is formed.

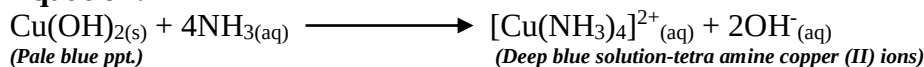
Ionically:



➤ **In excess ammonia:**

- The pale blue precipitate of $\text{Cu}(\text{OH})_{2(\text{s})}$ dissolves in excess ammonia to form a deep blue solution; proof that solution has Cu^{2+} ions;
- The deep blue solution is a complex salt of tetra-amine copper (II) ions.

Equation:





Uses of ammonia gas and its solution:

1. Ammonia gas is used in the manufacture of **nitric acid** and nylon.
2. Ammonia gas is important in the preparation of **ammonium salts** used as **fertilizers**.
3. It liquefies fairly easily (B.P is -33°C) and the liquid is used as a **refrigerant** in large cold storages and ice cream factories.
4. Liquid ammonia is injected directly into the soil as a high nitrogen content **fertilizer**.
5. Ammonia solution is used in **laundry work** as a **water softener** and a **cleansing agent** (stain remover)
6. Ammonia is used in the manufacture of sodium carbonate in the **Solvay process**.
7. Ammonia is used in “**smelling salts**”. It has a slightly stimulating effect on the action of the heart and so may prevent fainting

Qualitative analysis for cations using sodium hydroxide solution

(i). Procedure:

- To about 2cm^3 of solutions containing ions of calcium, magnesium, aluminium, zinc, iron, lead, copper etc in separate test tubes; aqueous sodium hydroxide is added dropwise till in excess.

(ii). Observations:

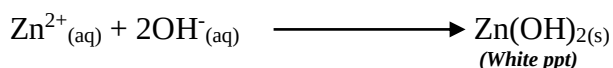
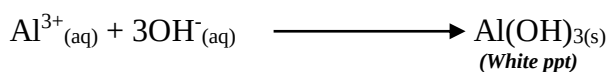
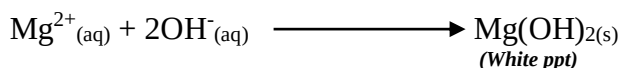
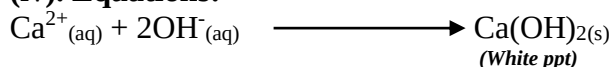
The various metal ions reacted as summarized in the table below.

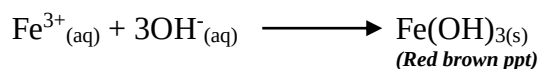
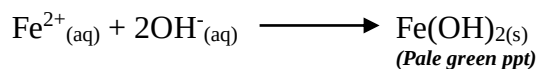
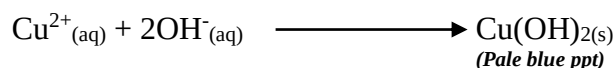
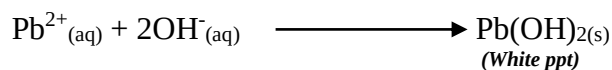
Metal ions in solution	Observations on addition of ammonia	
	Few drops of ammonia	Excess drops of ammonia
Ca^{2+}	White precipitate	White precipitate persists
Mg^{2+}	White precipitate	Precipitate persists;
Al^{3+}	White precipitate	Precipitate dissolves;
Zn^{2+}	White precipitate	Precipitate dissolves;
Fe^{2+}	Pale green precipitate	Precipitate persists; slowly turns red-brown on exposure to air;
Fe^{3+}	Red-brown precipitate	Precipitate persists;
Pb^{2+}	White precipitate	Precipitate dissolves;
Cu^{2+}	Pale blue precipitate	Precipitate dissolves forming a deep blue solution;

(iii). Explanations:

- Most metal ions in solution react with sodium hydroxide solution to form insoluble metal hydroxides.
- In excess sodium hydroxide, some of the so formed hydroxides (hydroxides of Zn, Al, Pb and Cu) dissolve forming complex ions.

(iv). Equations:



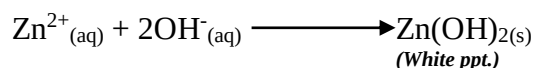


Note:

Hydroxides of $\text{Zn}^{2+}_{(\text{aq})}$; $\text{Pb}^{2+}_{(\text{aq})}$; and $\text{Al}^{3+}_{(\text{aq})}$ dissolve in excess ammonia solution forming complex ions.

1. Zinc ions and sodium hydroxide solution.

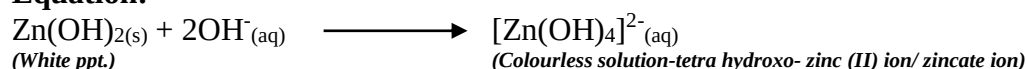
➤ **With little sodium hydroxide:**



➤ **In excess sodium hydroxide:**

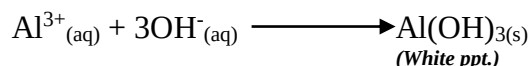
- The white precipitate of $\text{Zn}(\text{OH})_{2(\text{s})}$ dissolves in excess sodium hydroxide to form a colourless solution;
- The colourless solution is a complex salt of tetra-hydroxo zinc (II) ions (zincate ion).

Equation:



2. Aluminium ions and sodium hydroxide solution.

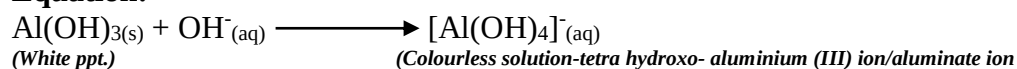
➤ **With little sodium hydroxide:**



➤ **In excess sodium hydroxide:**

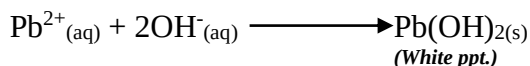
- The white precipitate of $\text{Al}(\text{OH})_{3(\text{s})}$ dissolves in excess sodium hydroxide to form a colourless solution;
- The colourless solution is a complex salt of tetra-hydroxo aluminium (III) ions (aluminate ion).

Equation:



3. Lead (II) ions and sodium hydroxide solution.

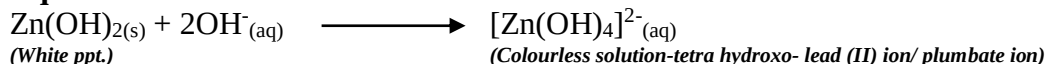
➤ **With little sodium hydroxide:**



➤ **In excess sodium hydroxide:**

- The white precipitate of $\text{Pb}(\text{OH})_{2(\text{s})}$ dissolves in excess sodium hydroxide to form a colourless solution;
- The colourless solution is a complex salt of tetra-hydroxo lead (II) ions (plumbate ions).

Equation:



Summary and useful information on qualitative analysis:
Colours of substances in solids and solutions in water.

COLOUR		IDENTITY
SOLID	AQUEOUS SOLUTION (IF SOLUBLE)	
1. White	Colourless	Compound of K^+ ; Na^+ ; Ca^{2+} ; Mg^{2+} ; Al^{3+} ; Zn^{2+} ; Pb^{2+} ; NH_4^+
2. Yellow	Insoluble	Zinc oxide, ZnO (turns white on cooling); Lead oxide, PbO (remains yellow on cooling, red when hot)
	Yellow	Potassium or sodium chromate;
3. Blue	Blue	Copper (II) compound, Cu^{2+}
4. Pale green Green	Pale green (almost colourless) Green	Iron (II) compounds, Fe^{2+} Nickel (II) compound, Ni^{2+} ; Chromium (II) compounds, Cr^{3+} ; (Sometimes copper (II) compound, Cu^{2+})
5. Brown	Brown (sometimes yellow)	Iron (III) compounds, Fe^{3+} ;
	Insoluble	Lead (IV) oxide, PbO_2
6. Pink	Pink (almost colourless) Insoluble	Manganese (II) compounds, Mn^{2+} ; Copper metal as element (sometimes brown but will turn black on heating in air)
7. Orange	Insoluble	Red lead, Pb_3O_4 (could also be mercury (II) oxide, HgO)
8. Black	Purple Brown Insoluble	Manganate (VII) ions (MnO_4^-) as in KMnO_4 ; Iodine (element)-purple vapour Manganese (IV) oxide, MnO_2 Copper (II) oxide, CuO Carbon powder (element) Various metal powders (elements)

Reactions of cations with common laboratory reagents and solubilities of some salts in water

CATION	SOLUBLE COMPOUNDS (IN WATER)	INSOLUBLE COMPOUNDS (IN WATER)	REACTION WITH AQUEOUS SODIUM HYDROXIDE	REACTION WITH AQUEOUS AMMONIA SOLUTION
Na^+	All	None	No reaction	No reaction
K^+	All	None	No reaction	No reaction
Ca^{2+}	Cl^- ; NO_3^- ;	CO_3^{2-} ; O^{2-} ; SO_4^{2-} ; OH^- ;	White precipitate insoluble in excess	White precipitate insoluble in excess, on standing;
Al^{3+}	Cl^- ; NO_3^- ; SO_4^{2-}	O^{2-} ; OH^- ;	White precipitate soluble in excess	White precipitate insoluble in excess
Pb^{2+}	NO_3^- ; ethanoate;	All others;	White precipitate soluble in excess	White precipitate insoluble in excess
Zn^{2+}	Cl^- ; NO_3^- ; SO_4^{2-}	CO_3^{2-} ; O^{2-} ; SO_4^{2-} ; OH^- ;	White precipitate soluble	White precipitate soluble



			in excess	in excess
Fe^{2+}	$\text{Cl}^-; \text{NO}_3^-; \text{SO}_4^{2-}$	$\text{CO}_3^{2-}; \text{O}^{2-}; \text{OH}^-;$	(Dark) pale green precipitate insoluble in excess	(Dark) pale green precipitate insoluble in excess
Fe^{3+}	$\text{Cl}^-; \text{NO}_3^-; \text{SO}_4^{2-}$	$\text{CO}_3^{2-}; \text{O}^{2-}; \text{OH}^-;$	(Red) brown precipitate insoluble in excess	(Red) brown precipitate insoluble in excess
Cu^{2+}	$\text{Cl}^-; \text{NO}_3^-; \text{SO}_4^{2-}$	$\text{CO}_3^{2-}; \text{O}^{2-}; \text{OH}^-;$	Pale blue precipitate insoluble in excess	Pale blue precipitate soluble in excess forming a deep blue solution
NH_4^+	All	None;	Ammonias gas on warming	Not applicable.

Qualitative analysis for common anions.

	$\text{SO}_4^{2-}(\text{aq})$	$\text{Cl}^-(\text{aq})$	$\text{NO}_3^-(\text{aq})$	$\text{CO}_3^{2-}(\text{aq})$
TEST	Add $\text{Ba}^{2+}(\text{aq})$ ions from $\text{Ba}(\text{NO}_3)_2(\text{aq})$; acidify with dilute $\text{HNO}_3(\text{aq})$	Add $\text{Ag}^+(\text{aq})$ from $\text{AgNO}_3(\text{aq})$. Acidify with dilute HNO_3 Alternatively; Add Pb^{2+} from $\text{Pb}(\text{NO}_3)_2$ and warm	Add $\text{FeSO}_4(\text{aq})$; Tilt the tube and carefully add 1-2 cm^3 of concentrated $\text{H}_2\text{SO}_4(\text{aq})$	Add dilute $\text{HNO}_3(\text{aq})$; bubble gas through lime water;
OBSERVATION	The formation of a white precipitate shows presence of SO_4^{2-} ion;	The formation of a white precipitate shows presence of Cl^- ion; Formation of a white precipitate that dissolves on warming shown presence of $\text{Cl}^-(\text{aq})$ ions	The formation of a brown ring shows the presence of NO_3^- ions	Evolution of a colourless gas that forma a white precipitate with lime water, turns moist blue litmus paper red; and extinguishes a glowing splint shows presence of CO_3^{2-} ions
EXPLANATION	Only BaSO_4 and BaCO_3 can be formed as white precipitates. BaCO_3 is soluble in dilute acids and so BaSO_4 will remain on adding dilute nitric acid	Only AgCl and AgCO_3 can be formed as white precipitates. AgCO_3 is soluble in dilute acids but AgCl is not; - PbCl_2 is the only white precipitate that dissolves on warming	Concentrated H_2SO_4 forms nitrogen (II) oxide with $\text{NO}_3^-(\text{aq})$ and this forms brown ring complex ($\text{FeSO}_4 \cdot \text{NO}$) with FeSO_4 ;	All CO_3^{2-} or HCO_3^- will liberate carbon (IV) oxide with dilute acids

Checklist:

1. Why is it not possible to use dilute sulphuric acid in the test for SO_4^{2-} ions;
2. Why is it not possible to use dilute hydrochloric acid in the test for chloride ions?
3. Why is it best to use dilute nitric acid instead of the other two mineral acids in the test for CO_3^{2-} ions?
4. How would you distinguish two white solids, Na_2CO_3 and NaHCO_3 ?

What to look for when a substance is heated.

1. Sublimation	White solids on cool, parts of a test tube indicates NH_4^+ compounds; Purple vapour condensing to black solid indicates iodine crystals;
2. Water vapour (condensed)	Colourless droplets on cool parts of the test tube indicate water of crystallization or HCO_3^- (see below)
3. Carbon (IV) oxide	CO_3^{2-} of Zn^{2+} ; Pb^{2+} ; Fe^{2+} ; Fe^{3+} ; Cu^{2+} ;



4. Carbon (IV) oxide and water vapour (condensed)	HCO_3^-
5. Nitrogen (IV) oxide	NO_3^- of Cu^{2+} ; Al^{3+} ; Zn^{2+} ; Pb^{2+} ; Fe^{2+} ; Fe^{3+}
6. Oxygen	NO_3^- or BaO_2 ; MnO_2 ; PbO_2 ;

Industrial manufacture of ammonia-The Haber process.

- Most of the world's supply of ammonia is from the synthesis of Nitrogen and hydrogen in the Haber process.

(i). Raw materials

➤ Nitrogen

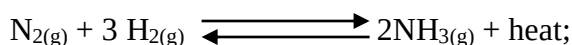
- Usually obtained from liquid air by fractional distillation

➤ Hydrogen

- Obtained from water gas in the **Bosch process**.

- Also from crude oil (cracking)

(ii). General equation



Note:

- Nitrogen and hydrogen combine in the ratio 1:3 respectively to form two volumes of ammonia gas plus heat.

-The reaction is **exothermic** releasing heat to the surrounding.

(iii). Conditions

➤ High pressures

- The process is favoured by high pressures and thus a pressure of approximately 200 to 300 atmospheres is used.

Reason:

- The volume of gaseous reactants from equation is higher than volume of gaseous products. Thus increased pressure shifts the equilibrium to the right; favoring the production of more ammonia.

Note:

Such high pressures are however uneconomical.

➤ Low temperatures

- Low temperatures favour production of ammonia;

Reason:

- The reaction is exothermic (releases heat to the surrounding) hence lower temperature will favour the forward reaction (shift the equilibrium to the right), producing more ammonia.

➤ Catalyst



- The low temperatures make the reaction slow and therefore a catalyst is used to increase the rate of reaction
- The catalyst used is **finely divided iron**; impregnated with Aluminium oxide (Al_2O_3)

(iv). The chemical processes

Step 1: Purification

- The raw materials, nitrogen and hydrogen are passed through a purification chamber in which impurities are removed.
- The main impurities are CO_2 , water vapour, dust particles, SO_2 , CO_2 and O_2 ;

Reason:

The impurities would poison the catalyst

Step 2: Compression

- The purified Nitrogen and Hydrogen gases are compressed in a compressor at 500 atmospheres.

Reasons:

- To increase chances of molecules reacting;
- To increase rate of **collision** of the reacting particles.
- To increase pressure (attain desired pressures); and hence increase concentration of reacting particles.

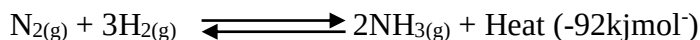
Step 3: Heat exchanger reactions

- Upon compression the gaseous mixture, nitrogen and hydrogen are channeled into a heat exchanger; which **heats them** increasing their temperature.
- This enables the reactants (hydrogen and nitrogen) to attain the **optimum temperatures** for the succeeding reactions (in the catalytic chamber)
- From the heat exchanger the gases go to the catalyst chamber.

Step 4: Catalytic chamber.

- The gases then combine in the ratio of 1:3 ($\text{N}_2:\text{H}_2$ respectively), to form ammonia.
- This reaction occurs in presence of a **catalyst**; which speeds up the rate of ammonia formation;
- The catalyst is **finely divided iron** impregnated with **aluminium oxide** (Al_2O_3 increases the catalytic activity of iron).

Equation in catalytic chamber



- Only about 6-10% of the gases combine.
- Due to the high heat evolution involved, the products are again taken back to the heat exchanger; to cool the gases coming from the catalytic chamber.

Step 5: Heat exchanger

- The gases from the catalytic chamber enter the heat exchanger a second time.

Reason:

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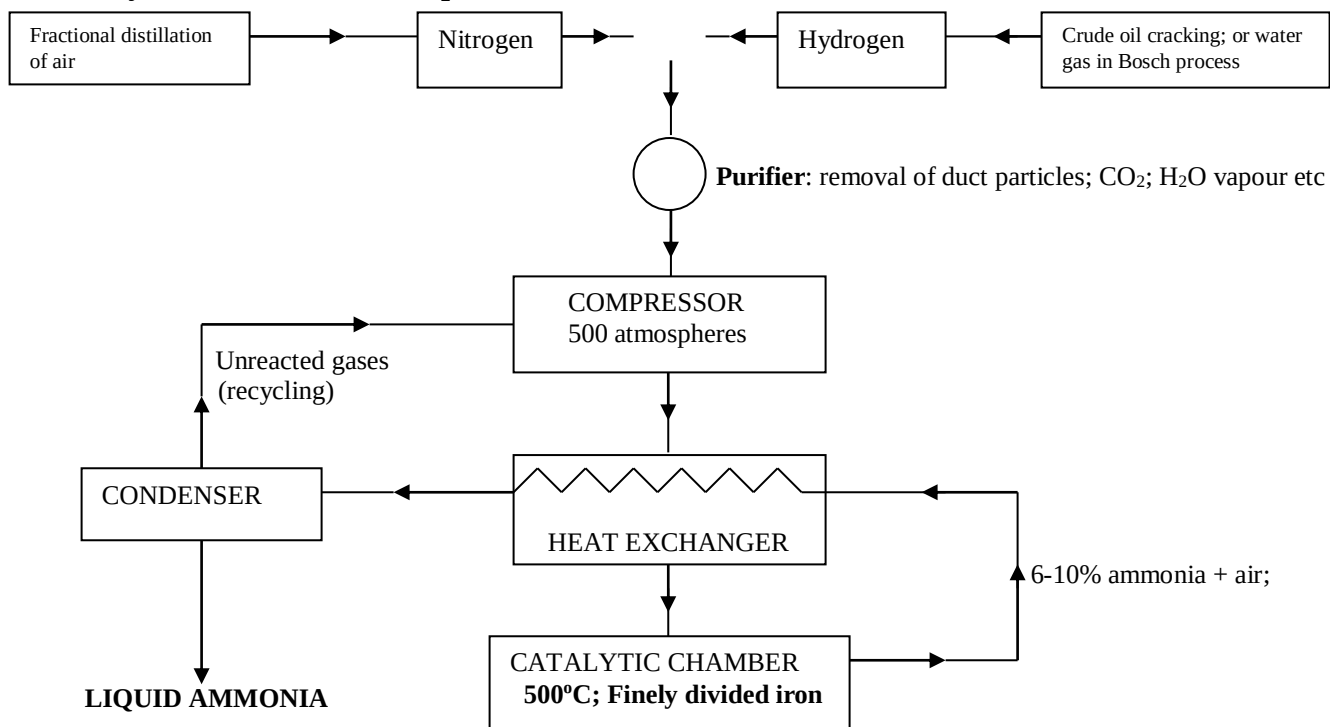


- To **cool** the gases coming from the catalytic chamber, thus reduce cost of condensation.
- The gaseous mixture; ammonia and uncombined nitrogen and hydrogen are the passed through a condenser.

Step 6: The condenser reactions (cooling chamber)

- The pressure and the low temperatures in this chamber liquefy ammonia, which is then drawn off.
- The uncombined (unreacted) gases are recirculated back to the compressor, from where they repeat the entire process.

Summary: flow chart of Haber process.



Citing a Haber process plant

- When choosing a site for this industrial plant, the following factors are considered:
 1. Availability of raw materials (natural gas and crude oil)
 2. Presence of cheap sources of energy.
 3. Availability of transport and marketing.
 4. Availability of appropriate technology and labour force.

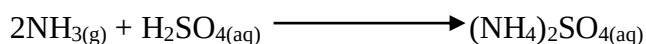
Ammonium salts as fertilizers

- Ammonium salts are prepared by the reaction between ammonia and the appropriate acid in dilute solution followed by evaporation and crystallization

(a). Ammonium sulphate

- Is prepared by absorbing ammonia in sulphuric acid.

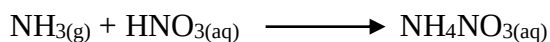
Equation:



Note: It is a cheap fertilizer.

(b). Ammonium nitrate

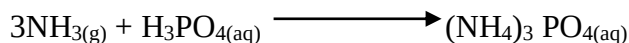
- Is prepared by neutralization nitric acid by ammonia.

Equation:

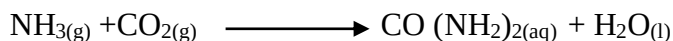
- As there is some danger of exploding during storage, ammonium nitrate is mixed with finely powdered limestone (CaCO_3).
- The mixture, sold as nitro-chalk is much safer.

(c). Ammonium phosphate

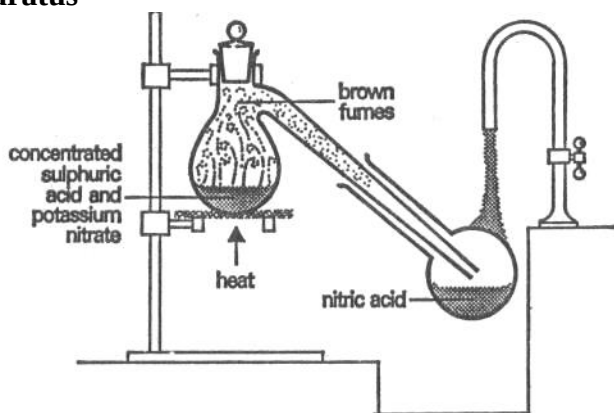
- It is particularly useful as it supplies both nitrogen and phosphorus to the soil.
- It is prepared by neutralizing **orthophosphoric** acid by ammonia

Equation:**(d) Urea $\text{CO}(\text{NH}_2)_2$**

- Is made from ammonia and carbon (IV) oxide
- Its nitrogen content by mass is very high; nearly 47%

Equation:**Nitric (V) acid**

- Is a monobasic acid (has only one replaceable Hydrogen atom); and has been known as strong water (aqua forty).
- It is a compound of hydrogen, oxygen and nitrogen.

Laboratory preparation of nitric (V) acid**(i). Apparatus****(ii). Reagents**

- Nitric acid is prepared in the laboratory by action of concentrated sulphuric acid on solid nitrates e.g. potassium nitrate (KNO_3) and sodium nitrate (NaNO_3)

(iii). Procedure

- 30-40 grams of small crystal of KNO_3 are put in a retort flask.
- Concentrated sulphuric acid is added just enough to cover the nitrate; and then heated (warmed) gently.
- The apparatus is all glass.

**Reason:**

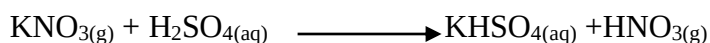
- Nitric (V) acid would attack rubber connections.
- The neck of the retort flask is inserted into a flask that is kept cool continually under running water; this is where nitric acid is collected.

Note:

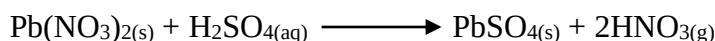
The cold water running over the collection flask is meant to cool (condense) the hot fumes of nitric (V) acid.

(iv). Observations and explanations

- Fumes of nitric are observed in the retort;

Equation

- If Lead (II) nitrate was used;

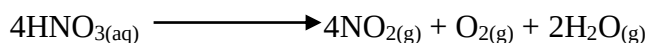


Note: with lead (II) nitrate the reaction soon stops because the insoluble lead (II) sulphate coats the surface of the nitrate preventing further reaction; yield of nitric (V) acid is thus lower;

- These fumes of nitric acid appear brown.

Reason:

- Due to the presence of nitrogen (iv) oxide gas formed by thermal decomposition of nitric acid.

Equation:

- Pure nitric (V) acid is colourless but may appear yellow (brown) due to the presence of Nitrogen (IV) oxide.
- The brown colour can be removed by blowing air through the acid.
- Fuming nitric acid boils at 83°C and is 99% pure; while concentrated nitric acid is only 70% acid and 30% water.

Note: Nitric acid is usually stored in dark bottles.

Reason:

- To avoid its decomposition by light to nitrogen (IV) oxide, oxygen and water.
- The reaction in the retort flask is a typical displacement reaction; in which the more volatile nitric (V) acid is displaced from nitrates by the less volatile sulphuric acid.
- The nitric acid distills over because it is more volatile than sulphuric acid.



Properties of concentrated nitric acid

- Nitric (V) acid readily gives oxygen and therefore is called an oxidizer.
- The acid is usually reduced to nitrogen (IV) oxide and water.

1. Effects of heat on concentrated nitric acid

(i) Apparatus

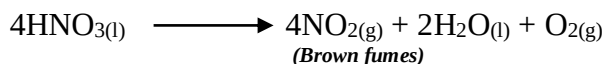
(ii) Observations

- Brown fumes are seen in the hard glass tube.
- Colourless gas is collected over water.

(ii). Explanations

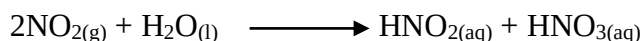
- Sand soaked in concentrated nitric acid produces nitric solid vapour on heating.
- The hot glass wool catalyzes the decomposition of nitric acid to nitrogen (IV) oxide (brown fumes), water vapour and oxygen.

Equation



- The so formed nitrogen (IV) oxide dissolves in water forming both nitric and nitrous acids.

Equation:



- The oxygen gas is collected over water; and with the solution becoming acidic.

2. Reaction with saw dust

- Saw dust contains compounds of carbon Hydrogen and oxygen.

Procedure:

- Some saw dust is heated in an evaporating dish and some few drops of concentrated nitric (V) acid on it (this is done in a fume cupboard)

Observation:

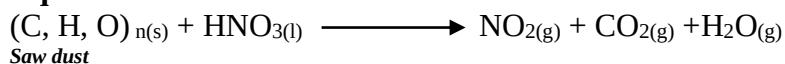
- A violent reaction occurs, the saw dust catches fire easily and a lot of brown fumes of nitrogen (IV) oxide given off.

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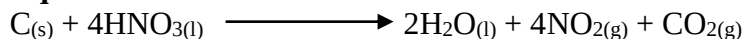
- Nitric (V) acid oxidizes the compounds in saw dust to CO_2 and water; and itself it is reduced to nitrogen (IV) oxide and water.

Equation:



- Warm concentrated nitric (V) acid oxidizes pure carbon and many other compounds containing carbon.

Equation:



3. Reaction with sulphur

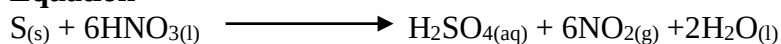
Procedure:

- 2 cm³ of concentrated nitric (V) acid is added to a little sulphur in a test tube and warmed.
- The mixture is filtered to remove excess sulphur and the filtrate diluted with distilled water.
- Drops of barium chloride are then added to the resultant solution.

Observations:

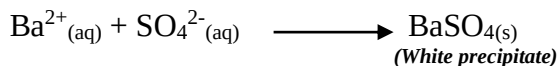
- Red brown gas, nitrogen (IV) oxide (NO_2) is evolved and the sulphur is oxidized to sulphuric acid.

Equation



- On addition of barium chloride to the solution, a white precipitate is formed.
- This is due to formation of barium sulphate and is a confirmation for the presence of SO_4^{2-} ions.

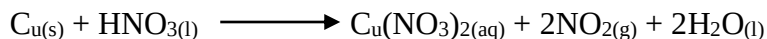
Equation:



4. Reaction with metals

- Concentrated nitric (V) acid reacts with metals except **gold** and **platinum**.
- Actual reaction depends on the concentration of the acid and the position of the metal in the reactivity series.
- The reaction results in a metal nitrate, NO_2 and water.
- Copper, which is low in the reactivity series, reduces conc. HNO_3 to NO_2 .

Equation:



- Metals more reactive than copper e.g. Magnesium may reduce nitric acid to dinitrogen monoxide (N_2O) or Nitrogen (N_2).
- Some metals like iron and aluminium form insoluble layers when reacted with nitric acid thus stopping any further reaction.

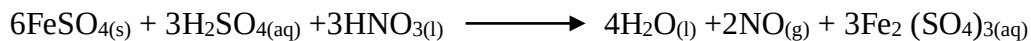
5. Reaction with iron (II) salts

**Procedure:**

- Few crystals of iron (II) sulphate are dissolved in dilute sulphuric acid.
- A little concentrated nitric (V) acid is added to the solution and mixture warmed.

Observation:

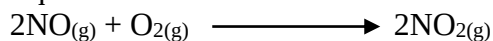
- Green solution turns brown.

Equation:**Explanation:**

- Nitric acid oxidizes green iron (II) salts (Fe^{2+}) to brown iron (III) salts (Fe^{3+}) and itself is reduced to Nitrogen (II) Oxide.

Note:

- In air, nitrogen (II) oxide is readily oxidized to nitrogen (IV) oxide; resulting to brown fumes.

Equation:**6. Reduction of nitric (V) acid by hydrogen sulphide.****Procedure**

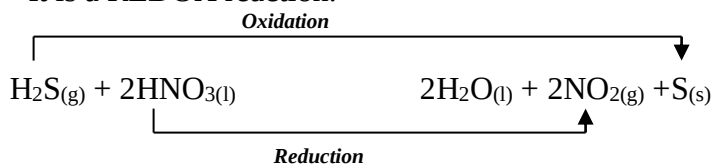
- A few drops of conc. nitric (V) acid are added to a gas jar full of hydrogen sulphide and the jar then covered.

Observations

- Fumes (brown) of Nitrogen (IV) oxide and yellow deposits of sulphur;

Equation

- It is a REDOX reaction.

**Properties of dilute nitric (V) acid****1. Reaction with metals**

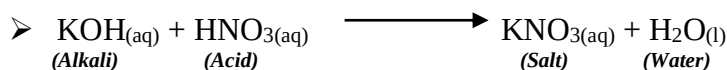
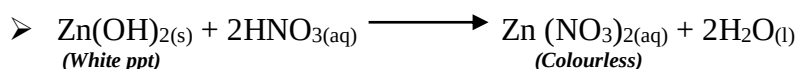
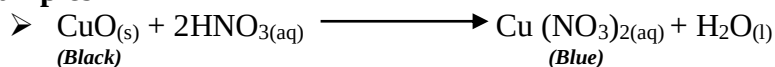
- Dilute nitric (V) acid reacts with most metals to form nitrogen (II) oxide instead of hydrogen.

**Example:**

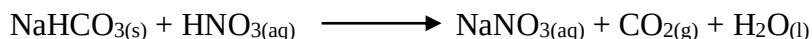
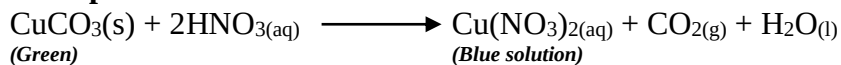
- In fact HNO_3 is reduced to NO and water but NO soon gets oxidized in air to form brown fumes of NO_2 .
- However very dilute HNO_3 (cold) reacts with more active metals such as Magnesium to produce **Hydrogen**.

2. Dilute nitric (V) acid as a typical acid

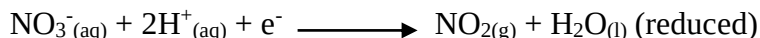
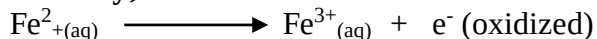
- It turns blue litmus paper red.
- It reacts with metal oxides and metal hydroxides to form a metal nitrate and water (Neutralization)

Examples**3. Reaction with metal carbonates and hydrogen carbonates**

- Dilute HNO_3 reacts with metal carbonates and hydrogen carbonates to form a nitrate, CO_2 and water.

Examples.**Test for nitrates/nitric acid****1. Oxidation of iron (ii) sulphate**

- Concentrated HNO_3 oxidizes green Iron (II) sulphate in presence of sulphuric acid into Iron (III) sulphate (yellow/brown)
- However the solution turns dark brown due to formation of a compound, $\text{FeSO}_4 \cdot \text{NO}$
- NO is produced by reduction of nitrate to nitrogen monoxide by Fe^{2+}

Ionically;**2. Brown ring test****Procedure.**

- An unknown solid is dissolved then acidified using dilute H_2SO_4 .
- Some FeSO_4 solution is then added.



- The test tube is then held at an angle and concentrated sulphuric (V) acid is added slowly (dropwise) to the mixture.

Observations

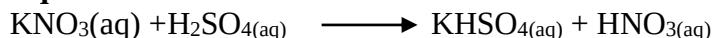
- The oily liquid (conc. H_2SO_4) is denser than water hence sinks to the bottom.
- A brown ring forms between the two liquid layers if the solid is a nitrate.

Diagrams:

Explanations:

- Suppose the solution tested is KNO_3 , the conc. H_2SO_4 and the KNO_3 reacted to produce HNO_3 .

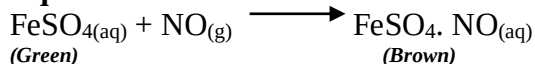
Equation:



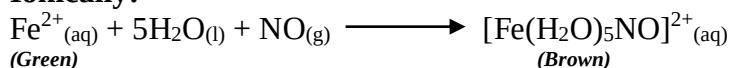
- The NO_3^- from nitric acid oxidizes some of the FeSO_4 to $\text{Fe}_2(\text{SO}_4)_3$ (Fe^{2+} to Fe^{3+}) and itself reduced to NO by the Fe^{2+}

-The NO so formed reacts with more FeSO_4 to give a brown compound ($\text{FeSO}_4 \cdot \text{NO}$) which appears as a brown ring.

Equation:



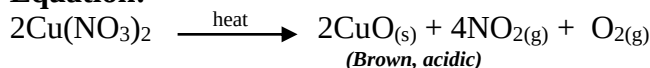
Ionic:



3. Heat

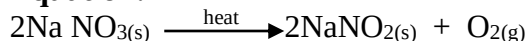
- Nitrates of less reactive metals decompose easily with gentle heating; clouds of brown NO_2 can be seen.

Equation:



- The nitrates of more reactive metals need much stronger heating and decompose in a different way.

Equation:



Uses of nitric acid

- Large quantities are used in fertilizer manufacture.
- Manufacture of explosives (TNT)
- Manufacture of dyes
- Making nitrate salts



- Etching of metals.
- Manufacture of nylon and terylene
- Refining precious metals
- An oxidizing agent.

Industrial manufacture of nitric acid

The Ostwald's process

(a). Introduction

- Nitric acid is manufactured by the catalyst oxidation of ammonia and dissolving the products in water.

(b). Raw materials

- Atmosphere air
- Ammonia from Haber process.

(c). Conditions

- **Platinum-rhodium** catalyst or platinum gauze.
- The ammonia-air mixture must be cleaned (purified) to remove dust particles which could otherwise poison the catalyst.

(d). Chemical reactions.

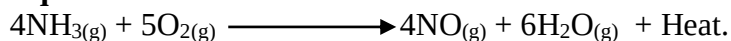
Step 1: Compressor reactions.

- Ammonia and excess air (oxygen) (1:10 by volume) is slightly compressed.
- The mixture is then cleaned to remove particles which would otherwise **poison** the catalyst.
- They are then passed to the heat exchanger.

Step 2: Heat exchanger and catalytic chamber.

- In the heat exchanger, the gaseous mixture is heated to about 900°C and then passed over a platinum-rhodium catalytic chamber.
- An **exothermic** reaction occurs and ammonia is oxidized to **nitrogen (II) oxide** and **steam**.

Equation:



- The exothermic reaction once started, provides the **heat** necessary to maintain the required **catalytic temperature**.
- This is of **economical advantage** i.e. electrical heating of catalyst is not continued hence lowering production costs.

Step 3: Heat exchanger.

- The hot products from catalytic chamber are again passed back through the heat exchanger.
- The hot gases are cooled and then passed into the cooling chamber.

Step 4: Cooling chamber

- Once cooled, the NO is oxidized to NO₂ by reacting it with excess oxygen.

Equation:





Step 5: Absorption towers:

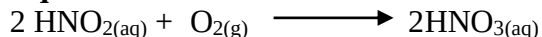
- The NO_2 in excess air is then passed through a series of absorption towers where they meet a stream of hot water and form nitric (V) acid and nitrous (III) acid.

Equations:



- The so produced nitrous (III) acid is oxidized by oxygen in excess air to nitric (V) acid so that the concentration of nitric acid in the solution (liquid) gradually increases.

Equation:



- The resultant HNO_3 is only 55%-65% concentrated.
- It is made more concentrated by careful distillation of the solution.

The process of distillation (increasing the concentration).

- Concentrated sulphuric (VI) acid is added to the dilute nitric (V) acid.
- The heat produced (when dilute sulphuric acid reacts with water) vapourises the nitric (V) acid.
- The resultant nitric (V) acid vapour is condensed.

Note:

- Nitric (V) acid is stored in dark bottles.

Reason:

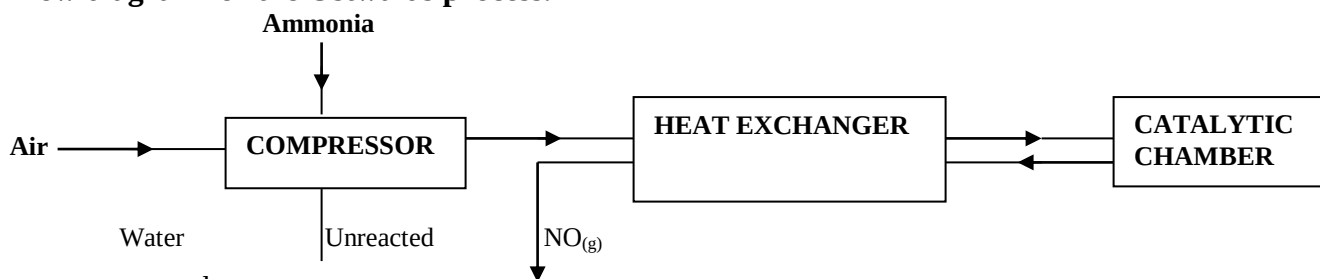
- To prevent its **decomposition** since it undergoes slow decomposition when exposed to light.

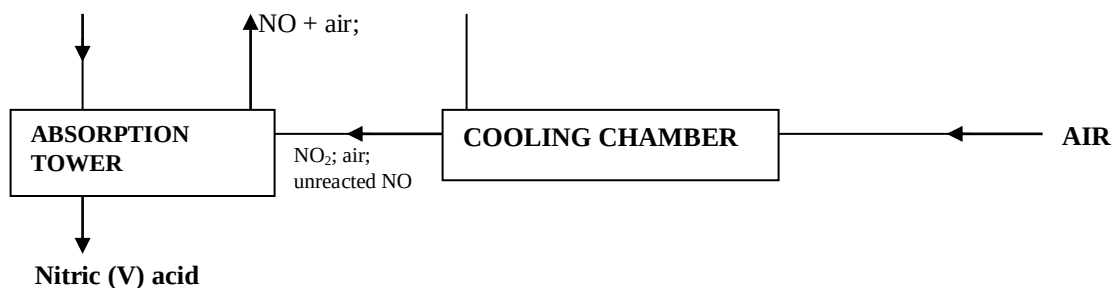
- Dilute nitric (V) acid has higher ions concentration than concentrated nitric (V) acid.

Reason.

- Dilute nitric (V) acid is a stronger acid hence ionizes fully to yield more hydrogen ions than concentrated nitric (V) acid.
- Dilute nitric (V) acid is ionic whereas concentrated nitric (V) acid is molecular;
- Dilute nitric (V) acid is more (highly) ionized than concentrated nitric (V) acid.

Flow diagram for the Ostwalds process.





Pollution effects of nitrogen compounds.

1. Acid rain

- Nitrogen (II) oxide is produced in internal combustion engines on combination of nitrogen and oxygen.
- Nitrogen (II) oxide oxidized to nitrogen (IV) oxide which dissolves in water to form nitric (III) and nitric (V) acids.
- Nitric (v) acid eventually reaches ground as acid rain and causes:
 - Loss of chlorophyll (chlorosis) from leaves
 - Corrosion of stone buildings and metallic structures, weakening them and destroying beauty.
 - Leaching of vital minerals from soils. These are converted into soluble nitrates and washed away from top soil. This leads to poor crop yields.

2. Smog formation.

- Nitrogen (IV) oxide also undergoes series of chemical reactions in air to produce one of the major components of **smog**.
- Smog reduces visibility for motorists, irritates eyes and causes breathing problems.

3. Eutrophication:

- Refers to enrichment of water with excess nutrients for algal growth.
- Presence of nitrate ions from nitrogen fertilizers in a water mass encourages rapid growth of algae.
- This eventually leads to reduction of dissolved oxygen in water, killing aquatic animals like fish.
- Presence of **nitrate ions** in **drinking water** may also cause **ill health** to humans. This is because they are converted into carcinogenic compounds.

Prevention.

1. Recycling unreacted gases in manufacture of nitric acid to prevent release into environment.
2. **Treating sewage** and **industrial effluents** to remove nitrogen compounds before releasing to rivers and lakes.
3. Fitting exhausts systems of vehicles with **catalytic converters** which convert nitrogen oxides into harmless nitrogen gas.
4. Adding lime to lakes and soils in surrounding regions to reduce acidity.
5. Applying fertilizers at right and in the correct proportion to prevent them from being washed into water masses.



UNIT 3: SULPHUR AND ITS COMPOUNDS

Checklist:

1. Occurrence of sulphur
2. Extraction of sulphur
 - The Frasch pump
 - Extraction process
3. Properties of sulphur
 - Physical properties
 - Chemical properties
4. Uses of sulphur
5. Allotropes of sulphur
 - Rhombic sulphur
 - Monoclinic sulphur
6. Compounds of sulphur
 - Sulphur (IV) oxide
 - Laboratory preparation
 - Other preparation methods
 - Properties of sulphur (IV) oxide
 - Physical properties
 - Chemical properties
 - Uses of sulphur (IV) oxide
7. Sulphur (VI) oxide
 - Laboratory preparation
 - Properties of sulphur (VI) oxide
8. Sulphuric (VI) acid
 - Large scale manufacture
 - Raw materials
 - The chemical process
 - Pollution control
 - Properties of concentrated sulphuric (VI) acid
 - Physical properties
 - Chemical properties
 - Properties of dilute sulphuric (VI) acid
 - Uses of sulphuric (VI) acid
8. Hydrogen sulphide gas
 - Laboratory preparation
 - Properties of hydrogen sulphide
 - Physical properties of hydrogen sulphide
 - Chemical properties of hydrogen sulphide
9. Atmospheric pollution by sulphur compounds

Occurrence

- Occurs naturally as a free element in the underground deposits in Texas and Louisiana (USA) and Sicily (ITALY).
- It also occurs as a sulphate and sulphide ores.

Examples;

Metallic sulphides: iron pyrites (FeS_2); Zinc blende (ZnS) Copper pyrites (CuFeS_2)

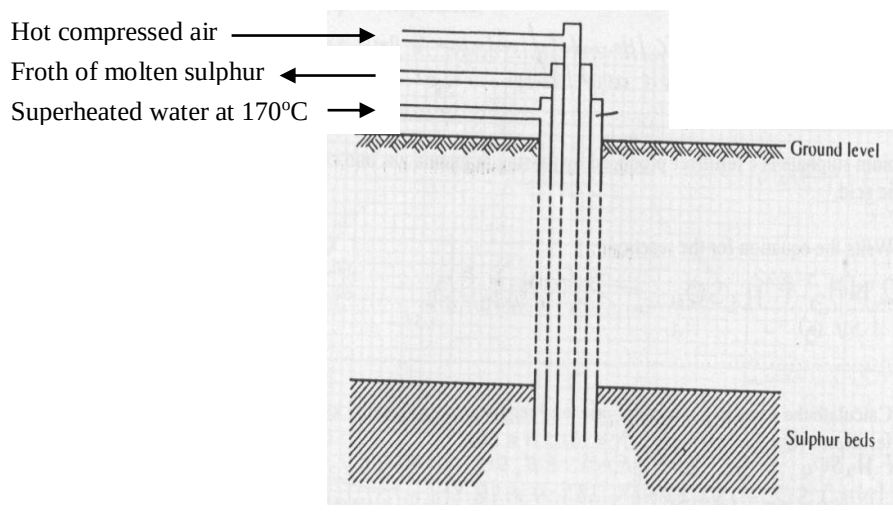
Metallic sulphates e.g. Gypsum, CaSO_4

Hydrogen sulphide e.g. H_2S present in natural gas.

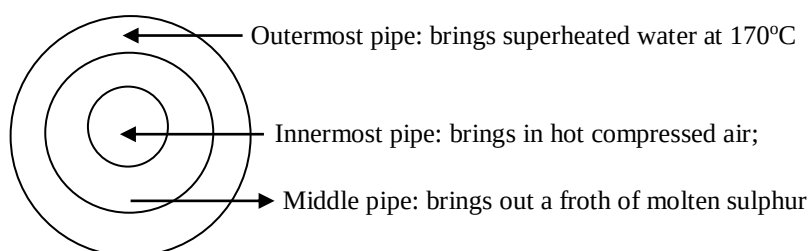
Extraction of sulphur: The Frasch process

- Is done using a set of 3 concentric pipes called **Frasch pump**; hence the name **Frasch process**.

(i). Apparatus: Frasch pump



Cross section of the Frasch pump



(ii). Chemical process

Note: Sulphur cannot be mined by conventional mining methods such as open cast, alluvial mining etc

Reasons:

- Sulphur deposits lie **very deep** under several layers of quicksand hence cannot be accessed easily.
- Sulphur deposits are associated with poisonous gases such as sulphur (IV) oxide gas which can cause massive pollution if exposed to open environment.
- Three concentric pipes, constituting the **Frasch pump** are drilled through the rock and soil down to the sulphur deposits.

(a). The outer tube (pipe)

- Is used to pump superheated water at 170°C and 10 atmospheres down the deposits.

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- The heat of the water melts the sulphur.
- By the time the water reaches the sulphur, its temperature drops to 120°C, but this is enough to melt sulphur whose M.P is 114°C.

(b). The innermost tube

- Is the smallest pipe and is used to blow or force a jet of hot compressed air down the sulphur deposits.
- This produces a light froth of molten sulphur (mixture of air, water and sulphur) which is forced up the middle pipe.

(c). The middle pipe.

- Allows the sulphur froth (mixture of molten sulphur, water and air) into the surface; where mixture is run into large tanks.
- The froth usually settles in two layers, the bottom layer is mainly water while the upper layer is mainly molten sulphur; due to differences in density.
- Once in the settling tanks, sulphur solidifies and separates out; giving 99% pure sulphur.
- The sulphur is removed, melted again and poured into moulds, to form roll sulphur in which form it is sold.

Properties of sulphur

Physical properties

1. - It is a yellow solid which exists in one amorphous form and 2 crystalline forms.
- A molecule of sulphur consists of a pluckered ring of 8 sulphur atoms covalently bonded.

Diagram: structure of a sulphur molecule.

2. Solubility

- It is insoluble in water but soluble in organic solvents like carbon disulphide, xylene and toluene.

3. It is a poor conductor of heat and electricity since it is a covalent element lacking free electrons or ions.

4. Effects of heat

- When sulphur is heated out of contact with air, it melts at low temperatures of about 113°C to form an **amber (orange) coloured** mobile liquid.

Reason:

- The S₈ rings open up to form chains of S₈.

Diagrams:

The pluckered S₈ ring of sulphur molecule

Chains of S₈ molecule



- On further heating, the liquid **darkens in colour**.
- At 160°C, the liquid becomes **much darker and very viscous** (such that the test tube can be inverted without the sulphur pouring out.)
- The viscosity continues to increase until a temperature of about 195°C

Reason:

- The S₈ rings of sulphur are **broken** and they then **join** to form **very long chains** of sulphur atoms, with over **100,000 atoms** (S_{100 000}).

Note: As the **chains entangle** with one another the **viscosity increases** and **colour darkens**.

- Near the **boiling point**, the liquid becomes **less dark** i.e. red-brown and more mobile (runny).

Reason

- The long chains are broken to shorter chains.

- At 444°C (boiling point), **sulphur vapourises** to form a **red-brown vapour** consisting of S₈, S₆, S₄ and S₂ molecules.

Reason

- The sulphur liquid changes state to form sulphur vapour.
- The vapour is **light brown** in colour, and consists of a mixture of molecules of formula S₂-S₁₀

Note

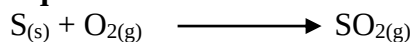
- If heated further the larger sulphur vapour molecules (S₈, S₆ etc) dissociate and at 750°C the vapour is mostly constituted of diatomic molecules (S₂)
- On exposure to **cold surfaces** the **light brown vapour** condenses to a **yellow sublimate**. The yellow sublimate is called flowers of sulphur.

Chemical properties

1. Burning in air

- It burns in air with a **bright blue flame** forming a misty gas with a choking smell.
- The gas is sulphur (IV) oxide, with traces of sulphur (VI) oxide, both of which are acidic.

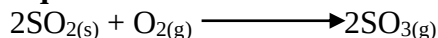
Equation:



Note:

The SO₃ is formed due to further oxidation of some of the SO₂ gas

Equation:



2. Reaction with acids.

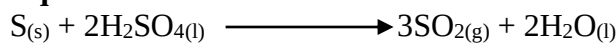
- Dilute acids have no effect on sulphur.
- It is however easily oxidized by concentrated (VI) sulphuric acid and Nitric (VI) acid.

➤ **With conc. H₂SO₄**



- When warmed with conc. H_2SO_4 , sulphur is oxidized to sulphur (IV) oxide while the acid is reduced to the same gas.

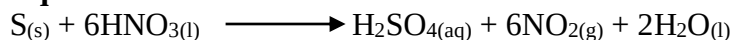
Equation:



➤ **With conc. HNO_3**

- Sulphur is oxidized to sulphuric (VI) acid while acid itself is reduced to red-brown Nitrogen (IV) oxide.

Equation:



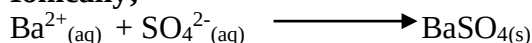
Note:

- The resultant solution gives a white precipitate with a solution of Barium chloride.

Reason

- Due to presence of sulphate ions which combine with Ba^{2+} to form insoluble $\text{BaSO}_{4(\text{s})}$

Ionically;



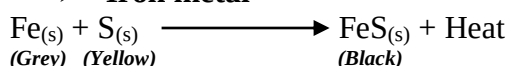
3. Reaction with other elements.

- It combines directly with many other elements to form sulphides.
- With metals, sulphur forms metal sulphides, most of which are black.

Examples.

(a). With metals

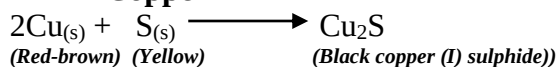
➤ **Iron metal**



Note:

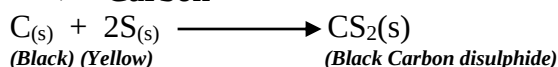
- During the reaction, the mixture glows spontaneously; immediately the reaction has started.

➤ **Copper**



(b). Non-metals

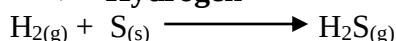
➤ **Carbon**



Note.

- Carbon (IV) sulphide has a distinct smell.
- It is an excellent solvent and is used as a **pesticide** due to its poisonous nature.

➤ **Hydrogen**

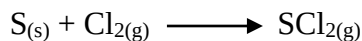




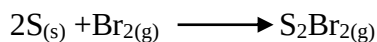
➤ **Fluorine**



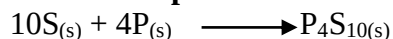
➤ **Chlorine**



➤ **Bromine**



➤ **Phosphorous**



Note:

- Sulphur does not react with inert gases, nitrogen and iodine.

Uses of sulphur

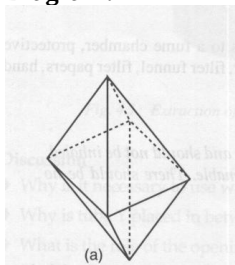
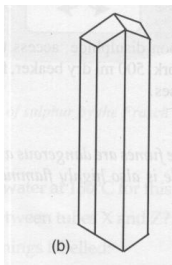
1. Industrial manufacture of sulphuric (VI) acid in the contact process.
2. It is used as a **fungicide** for treatment of fungal **skin diseases**.
3. It is used for **vulcanization** (hardening) of rubber
4. Manufacture of **calcium hydrogen sulphite** ($\text{Ca}(\text{HSO}_3)_2$) used for bleaching in paper and textile industries.
5. Manufacture of **matches** and **fireworks**.
6. Manufacture of **dyes** e.g. sulphur blacks that gives paint smooth texture.
7. Manufacture of sulphur **ointments** and **drugs** e.g. sulphur-guanidine for dysentery.
8. Manufacture of **hair oil**.
9. Small amounts of sulphur are added to **concrete** to prevent **corrosion by acids**.
10. Manufacture of fungicides for spraying crops against fungal infections e.g. ridomil, dithane for potato and tomato blights

Allotropes of sulphur

- Allotropy is the existence of an element in more than one form without change of state.
- Sulphur has 2 allotropes
 - Rhombic sulphur/ octahedral/ alpha-sulphur
 - Monoclinic/ prismatic sulphur/ beta-sulphur.

-Unlike carbon only the rhombic sulphur occurs naturally.

Comparison of rhombic and monoclinic sulphur.

Allotrope Characteristic	Rhombic sulphur	Monoclinic sulphur
Stability	- Stable below transitional temp. of 96°C	- Stable above 96°C
Colour	- Bright yellow crystalline solid	- Pale yellow crystalline solid
Melting point	- Melts at 113°C;	- Melts at 119°C;
Density	- About 2.06gcm ⁻³ (heavier than monoclinic Sulphur)	- Lighter than 1.98gcm ⁻³ (lighter than rhombic sulphur)
Shape	- Octahedral shape Diagram:  (a)	- Needle-like/ prismatic Diagram:  (b)

Note.

96°C is called transitional temperature; because both allotropes are stable.

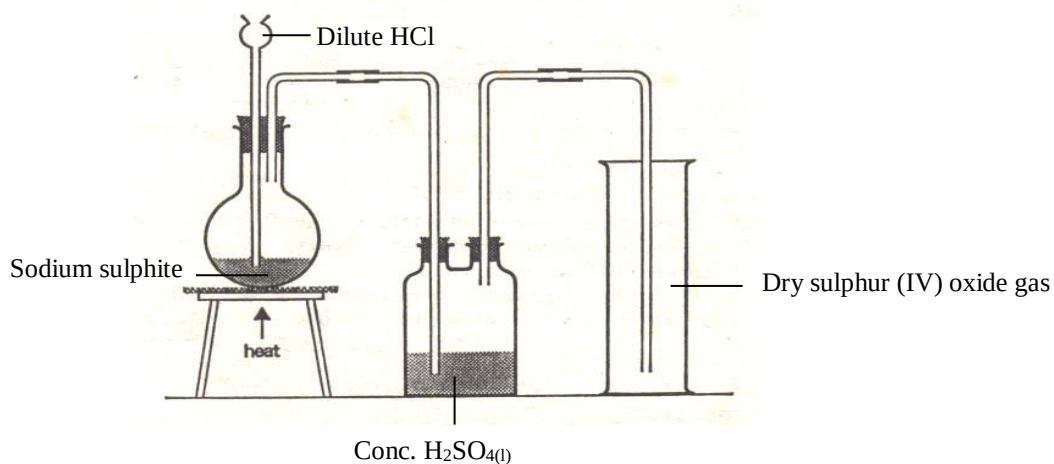
Compounds of sulphur

Oxides of sulphur.

Sulphur (IV) oxide

Laboratory preparation of sulphur (IV) oxide

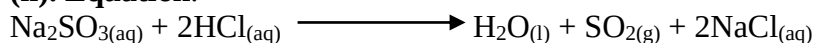
(i). Apparatus:

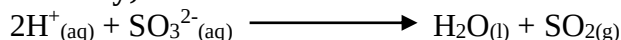


(ii). Procedure

- Dilute HCl or H₂SO₄ is poured into sodium sulphite crystals in the flask.
- The gas produced is passed through conc. Sulphuric acid to dry it.
- If the reaction is slow, the round-bottomed flask is heated (warmed) gently.
- Dry gas is collected by **downward delivery** as it is denser than air.

(ii). Equation.

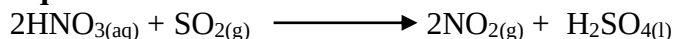


**Ionicall;****Note:**

- Nitric (V) acid should not be used.

Reason:

- It is a strong oxidizing agent and cannot therefore reduce the metal sulphites.
- Instead it will oxidize the SO_2 produced to sulphuric (VI) acid

Equation:**Other methods of preparing sulphur (IV) oxide.****(b). Preparation from concentrated sulphuric (VI) acid****(i). Apparatus**

- As in (a) above

(ii). Procedure

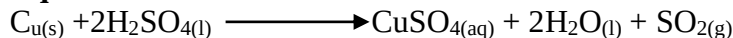
- Copper turnings are covered with concentrated sulphuric (VI) acid and the mixture **heated** (a must in this case).

Note:

- Dilute sulphuric (VI) acid does not react with copper hence the need for concentrated acid.
- Cold concentrated sulphuric (VI) acid does not also react with copper hence warming.

(iii). Observation.

- When the solution becomes hot, there is evolution of sulphur (IV) oxide gas.

Equation.**Note:**

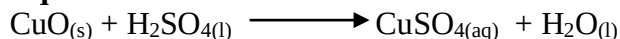
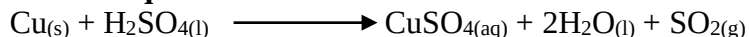
- This reaction is in two stages.

➤ **Oxidation of Cu to CuO**

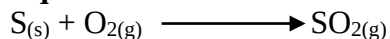
- Concentrated sulphuric (VI) acid oxidizes copper to Copper (II) oxide

Equation:

➤ **CuO further reacts with the acid to form salt and water.**

Equation:**Overall equation:****(c). Roasting sulphur in air**

- When sulphur is burnt in air, SO_2 is produced.

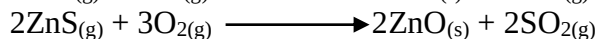
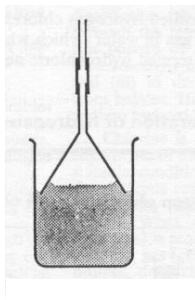
Equation:**Note:**

This reaction is not suitable for preparing a pure sample of the gas in the lab.

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Reason

- The gas is contaminated with traces of O₂; N₂; CO₂ and inert gases.
- There are higher chances of environmental pollution, due to escape of some of the gas into the atmosphere.

(d). Roasting metal sulphides in air**Examples:****Preparation of sulphur (IV) oxide solution.****(i). Apparatus****(ii). Procedure**

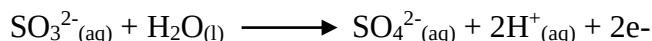
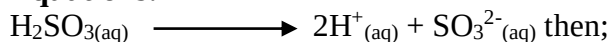
- Gas is directly passed into water using an **inverted funnel**; to prevent “**sucking back**” by increasing surface area for dissolution.

Properties of sulphur (IV) oxide gas**Physical properties**

1. It is a colourless gas with an irritating (pungent) characteristic smell.
2. It neither burns nor supports combustion i.e. when a lighted splint is introduced into a gas jar full of sulphur (IV) oxide, the splint is extinguished.
3. It has a low PH.

Chemical properties.

- It is a strong reducing agent.
- An aqueous solution of sulphur (IV) oxide, sulphurous acid is strong reducing agent.
- The sulphite radical, SO₃²⁻, acts as a supplier of electrons; the overall reaction results into formation of sulphate ions.

Equations:

- The resultant electrons supplied are accepted by an oxidizing agent, which consequently gets reduced.



Examples:

(i). Reduction of acidified potassium manganate (VII).

Procedure.

- To about 2 cm³ of sulphur (IV) oxide solution, 2 cm³ of dilute H₂SO₄ was added followed by an equal volume of potassium manganate (VII) solution.

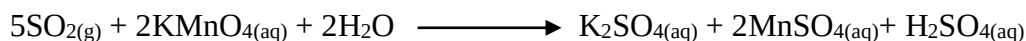
Observations

- Purple solution changes to colourless.

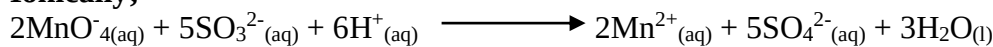
Explanation

- Purple manganate (VII) ions are reduced to colourless manganate (II) ions, while H₂SO₃ (sulphurous (IV) acid) is reduced to sulphate ions and water.

Equation:



Ionicall;



(ii). Reduction of potassium chromate (IV) solution

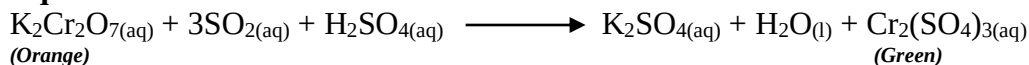
Procedure

- To 2 cm³ of Sulphur (IV) oxide solution, 2 cm³ of dilute H₂SO₄ was added followed by an equivalent volume of potassium chromate (VI) solution.

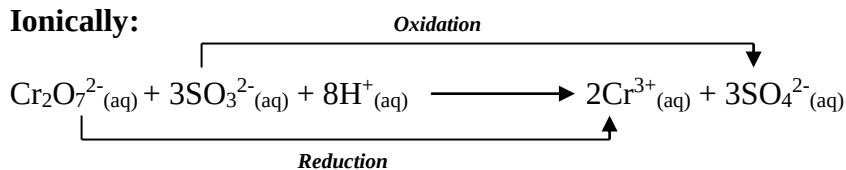
Observation

- Acidified potassium chromate (VI) solution change from **orange** to **green**.

Equation



Ionicall;



Note: this is the usual **chemical test** for sulphur (IV) oxide.

(iii). Reduction of Iron (III) ions to Iron (II) ions (Fe³⁺ to Fe²⁺)

Procedure

- About 3 cm³ of Iron (III) chloride solution are heated in a test tube and sulphur (IV) oxide gas bubbled into it.

Observations

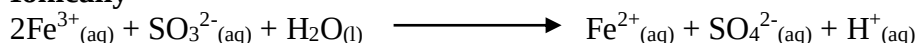
- The **brown** solution turns **green**.

Explanation

- Aqueous sulphur (IV) oxide reduces to Fe³⁺ in FeCl₃ which are brown to green Fe²⁺ in FeCl_{2(aq)}.



Ionic

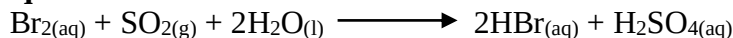


(iv). Reduction of bromine water

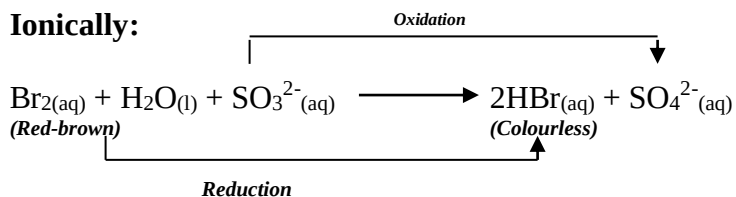
Procedure

- Bromine water (red brown) is added to a solution of sulphur (IV) oxide followed by HCl and BaCl₂ solution.

Equation



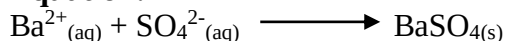
Ionic:



On addition of barium chloride

- A **white precipitate** is formed, due to the formation of insoluble **barium sulphate**.

Equation:



Note

- This test confirms presence of SO₄²⁻ since a white precipitate insoluble in dilute hydrochloric acid is formed.
- CO₃²⁻_(aq) and SO₃²⁻ also forms a white precipitate with BaCl_{2(aq)} but the white precipitates dissolve in dilute HCl_(aq)

(v). Reduction of hydrogen peroxide

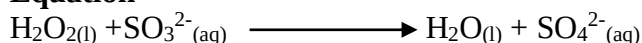
Procedure

- To 2 cm³ of aqueous sulphur (IV) oxide, an equal volume of hydrogen peroxide is added followed by 1 cm³ of HCl, then a few drops BaCl₂ solution.

Observation and explanations:

- Bubbles of a colourless gas; that relights a glowing splint.
- Hydrogen peroxide is reduced to water; while the sulphite ion in aqueous sulphur (IV) oxide (H₂SO_{3(aq)}) is oxidized to SO₄²⁻_(aq)

Equation

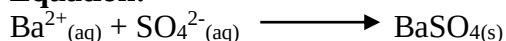


- On addition of BaCl₂, a white precipitate insoluble in dilute HCl.



- This confirms presence of sulphate ions.

Equation:



(vi). Reduction of concentrated nitric (V) acid

Procedure

- Sulphur (IV) oxide is bubbled through (into) a solution of concentrated nitric (v) acid.

Observation

- Brown fumes (of NO_2) are liberated.

Explanation

- Sulphur (IV) oxide reduces nitric (V) acid to nitrogen (IV) oxide (brown) while it is itself oxidized by HNO_3 to form H_2SO_4 .

- Thus while SO_2 is the reducing agent; HNO_3 is the oxidizing agent.

Equation:



(vii). Reaction with atmospheric oxygen in light.

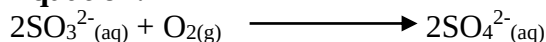
Procedure:

- About 2 cm^3 of Sulphur (IV) oxide solution is left in a test tube in light for 24 hours, dilute HCl is then added, followed by barium chloride.

Observations and explanations:

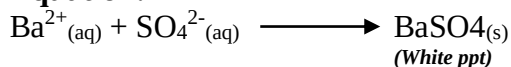
- Atmospheric oxygen in light oxidizes sulphite ion (SO_3^{2-}) into sulphate (SO_4^{2-})

Equation:



- On adding barium chloride, a **white precipitate** insoluble in dilute HCl results; confirming presence of sulphate ion.

Equation:



6. Sulphur (IV) oxide as oxidizing agent

- It reacts as an oxidizing agent with reducing agents more powerful than itself.

Examples

(a). Reaction with hydrogen sulphide

Procedure

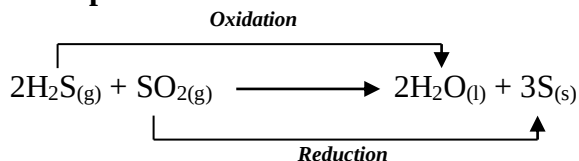
- A test tube of dry hydrogen sulphide gas is inverted into a gas jar full of **moist** sulphur (IV) oxide, and the gases allowed to mix.

Observation

- **Yellow** deposits of sulphur is produced.



Examples:



Explanations:

- H₂S is a **stronger reducing** agent than sulphur (IV) oxide.
- Thus sulphur (IV) oxide acts as an **oxidizing agent** supplying oxygen to the hydrogen sulphide.

Note

- Dry gases do not react and for this reaction to occur, the gases must be **moist** or at least one of them.

(b). Reaction with burning magnesium

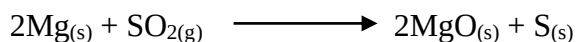
Procedure

- Burning magnesium is lowered into a gas jar full of sulphur (IV) oxide.

Observation

- **White fumes** of magnesium oxide and **yellow specks** of sulphur.

Equation



7. Sulphur (IV) oxide as bleaching agent.

Procedure

- Coloured flower petals are placed in a test-tube full of sulphur (IV) oxide.

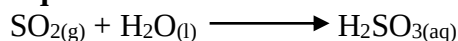
Observation

- The coloured (blue or red) petals are bleached (turned colorless);

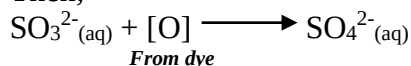
Explanations:

- In presence of water, sulphur (IV) oxide acts as a bleaching agent. It bleaches by reduction (removal of oxygen from the dye)
- It first combines with water forming the sulphurous acid; which then **reduces** the dye to form a colourless product.

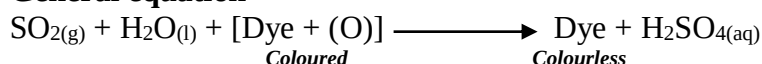
Equations:



Then;



General equation





Note

- The original colour may be restored by **oxidation** or prolonged exposure to air. This explains why old newspapers which were originally bleached white by sulphur (IV) oxide turn brown with time.
- Chlorine bleaches by oxidation hence its oxidation is **permanent**; SO₂ is however preferred because it is milder in action.

8. Reaction with sodium hydroxide (alkalis)

Procedure

- A gas jar full of sulphur (IV) oxide is inverted over sodium hydroxide solution in a trough and shaken.

Observations

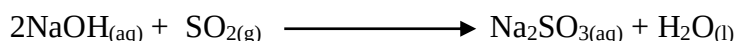
- Solution seen **rises up** in the jar.

Explanation

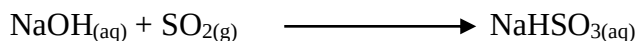
- Sulphur (IV) oxide is acidic, hence easily absorbed by alkaline solutions such as sodium hydroxide solution.
- Sodium sulphite and sodium hydrogen sulphites are formed depending on amount of sulphur oxide.

Equations

- **With limited sulphur (IV) oxide:**



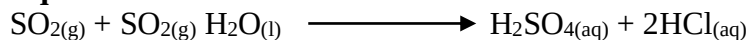
- **With excess sulphur (IV) oxide:**



Reaction with chlorine:

- Sulphur (IV) oxide reacts with moist chlorine to form an acidic mixture of sulphuric (VI) acid and hydrochloric acid.

Equation:



Explanation:

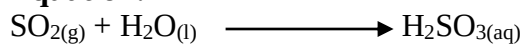
- Sulphur (IV) oxide serves as the reducing agent reducing chlorine into hydrochloric acid;
- Chlorine acts as the oxidizing agent; oxidizing the sulphur (IV) oxide into sulphuric (VI) acid

Tests for sulphur (iv) oxide

1. Characteristic pungent smell.
2. Bleaches flower petals.
3. Decolourises purple potassium manganate (VII)
4. Turns filter paper soaked in acidified orange potassium dichromate (VI) solution to green

Sulphur (IV) oxide as a pollutant

- It is industrial waste in some chemical processes.
- The emission to the air it dissolves forming sulphurous acid.

**Equation:**

- Sulphurous acid is readily oxidized to sulphuric (VI) acid; which attacks stonework and metal structures causing them to corrode.
- If breathed in, SO_2 causes lung damage.

Uses of sulphur (VI) oxide

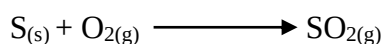
- Industrial manufacture of sulphuric (VI) acid.
- Fumigation in green houses for purposes of pest and disease control.
- Preservative in jam and fruit juices.
- Bleaching agent for wool, straw, paper pulp etc.

Sulphuric (VI) acid**Industrial manufacture of sulphuric (VI) acid: The contact process****Raw materials**

- Sulphide ores or sulphur.
- Water
- Oxygen (air)
- Concentrated sulphuric (VI) acid.

The chemical process**Step 1: Production of sulphur (IV) oxide**

- Sulphur (IV) oxide is obtained by burning the metal ores of sulphides or elemental sulphur in air.

Equation:

- Obtaining sulphur (IV) oxide from pyrites is **cheaper** than from sulphur.
- Flows of sulphur from pyrites is **impure** and contains dust; which involves extra expenses and time in purification.



Step 2: Purification and drying

- The Sulphur (IV) oxide and excess air are passed through a series of **driers** and **purifiers**.
- Purifiers remove dust particles, which would otherwise **poison** the catalyst used in this process by taking up the catalytic surface thus impairing the catalytic efficiency.
- Purification (removal of dust) is by **electrostatic precipitation**.
- Are dried through concentrated sulphuric acid then passed through heat exchanger.

Step 3: Heat exchanger reactions

- The pure dry SO₂ and excess air mixture are passed into heat exchanger reactions.

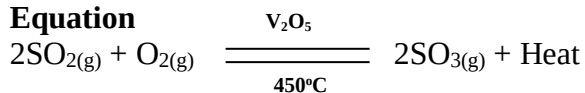
Reason:

- To **lower** their temperatures since reaction in the proceeding chamber (catalytic chamber) are **exothermic** hence requiring lower temperatures.

Step 4: Catalytic chamber

- Dry dust-free SO₂ is mixed with clean excess air, heated and passed into a catalytic chamber containing **vanadium (V) oxide catalyst**.

Equation



- The product is **sulphur (VI) oxide**, SO₃.
- Formation of sulphur (VI) oxide is accompanied by evolution of heat (**exothermic reaction**) and a reduction in volume.

Note:

- A good yield of SO₃ is favoured by the following conditions.

1. Temperature

- The forward reaction is **exothermic** hence the yield can be favourable in **low temperatures**.
- However, at such low temperatures the equilibrium is attained very slowly.
- At high temperatures, equilibrium is achieved very quickly but sulphur (VI) oxide decomposes considerably.
- Thus a compromise optimum temperature of about 450°C is used in order to enable as much sulphur (VI) oxide as possible to be made in a reasonable time.
- From the graph, high SO₃ yield is favoured by relatively low temperatures.

Graph: %age yield of sulphur (VI) oxide against temperature.



2. Pressure

- High pressures favour production of more sulphur (VI) oxide.

Reason

- The volume of **gaseous reactants** is higher than volume of **gaseous products**.
- Since reaction involves reduction in volume, theoretically pressure used should be as high as is economically convenient.

Note:

- High pressures are however disadvantageous.

Reason

- The equipment required to generate high pressure would be expensive to maintain.
- The high pressure could also liquefy sulphur (VI) oxide.
- A pressure slightly above atmospheric pressure is used providing 98% conversion at low maintenance costs.

3. Catalyst

- A catalyst neither takes part in a reaction nor increases the yield.
- It merely speeds up the reaction i.e. reduces the time taken to react at equilibrium of 450°C.
- Main catalyst is **vanadium (V) oxide** (V_2O_5).
- It is spread out (in trays) on silica gel to increase the surface area for combination of reactants.
- Dust settled in the catalyst may reduce its effective area.
- Dust may also react with the catalyst, "**poison**" it and further reduce its efficiency.
- This explains need to purify gases thoroughly.
- An effective catalyst is **platinised asbestos**.
- However, **vanadium (V) oxide** is preferred.

Reasons:

- It is **not easily poisoned** by dust particles.
- It is cheaper and readily available.

Note:

- The highest yield of sulphur (VI) oxide is obtained at optimum conditions of 450°C and pressure 2-3 atmospheres in presence of vanadium (V) oxide or platinised asbestos.

Step 5: Heat exchanger reactions

- Hot SO_3 gas from catalytic chamber is again passed through heat exchanger for **cooling** after which the cooled gas is taken into an absorption chamber.

Step 6: Absorption chamber

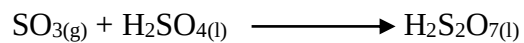
- The SO_3 is not dissolved (passed) into water directly.

Reason

- It dissolves in water **exothermically** with a loud, hissing sound giving off corrosive vapour resulting into harmful sulphuric acid "sprays" or mist all around.
- The SO_3 is dissolved in conc. H_2SO_4 forming oleum (pyrosulphuric acid/ fuming sulphuric acid).

Equation:

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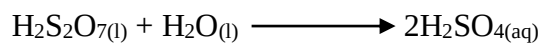


- Resultant “**Oleum**” is then channeled into a dilution chamber.

Step 7: Dilution chamber.

- Oleum is diluted with correct amounts of water to form concentrated sulphuric acid.

Equation:

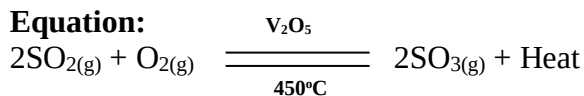


Summary: flow diagram for the contact process:



Pollution control in contact process.

- Main source of pollution is **sulphur (IV) oxide**.
- In catalyst chamber, SO_2 reacts with oxygen forming SO_3 .



- This is a reversible reaction and upto 98% conversion is possible and excess (unreacted) SO_2 warmed and released into atmosphere via long chimneys.
- However, SO_2 being a pollutant, little or none should be released into atmosphere.
- This is done by **scrubbing the gas**.
- This involves neutralizing the chimney gas by a solution of Calcium hydroxide forming a salt (calcium sulphite) and water.



Note:

- In certain cases, **filters** are also installed to remove any traces of acid spray or mist form the exhaust gases.
- The unreacted gases (SO_2 and SO_3) may also be recycled within the process.

Properties of concentrated sulphuric (VI) acid

Physical properties

1. - Colourless, odourless, oily liquid.
2. - Very dense; with density 1.84 gcm^{-3} .
3. - Soluble in water and gives out considerable heat when a solution is formed.
4. - It is hygroscopic – absorbs atmospheric moisture to become wet.

Experiment: To show hygroscopic nature of conc. H_2SO_4 .

(i). Procedure

- A small beaker half full of conc. H_2SO_4 is weighed.
- Level of acid in beaker is marked to the outside using gummed paper.
- Acid is left exposed to air for a week or so then weighed again and level also noted.

(ii). Observations

- There is an increase in weight of acid.
- Level of acid in beaker is now above the paper mark.

(iii). Explanations

- The increase in weight and size is due to water absorbed form the air by the conc. sulphuric (VI) acid.

Note:

- This explains why sulphuric (VI) acid is used as a **drying agent**.

Chemical properties

5. - It is a **dehydrating agent**.

Examples:

(a). Action on blue hydrated copper (II) sulphate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) crystals.

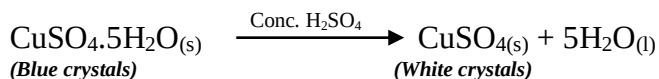
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**(i). Procedure**

- A few crystals of hydrated $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ were put in a test tube and enough concentrated sulphuric (VI) acid added, to cover them completely.

(ii). Observation:

- Blue copper (II) sulphate pentahydrate crystals turn to white powder of anhydrous CuSO_4 .

Equation**Explanations:**

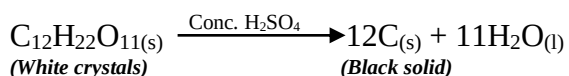
- Conc. H_2SO_4 has a very strong affinity for water and hence removes water of crystallization from crystals hence dehydrating them.

(b). Action on white sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$)**(i). Procedure:**

- A tablespoonful of sugar is put in an evaporating dish from a beaker and adequate volume of conc. H_2SO_4 is added.

(ii). Observations:

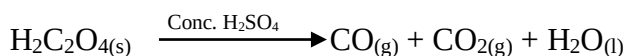
- Sugar turns from brown then **yellow** and finally to a **charred black mass of carbon**.
- A spongy black mass of charcoal (carbon) rises almost filling the dish.
- Steam is also given off and dish becomes very hot since reaction is exothermic.

Equation**Explanation**

- The acid removed from the sugar elements of water (hydrogen and oxygen, ratio 2:1) to form water, leaving behind a black charred mass of carbon.

(c). Action on oxalic acid (ethanedioic acid ($\text{H}_2\text{C}_2\text{O}_4$))

- Conc. H_2SO_4 dehydrates oxalic acid on heating to a mixture of carbon (II) oxide and carbon (IV) oxide.

Equation**Note:**

- Conc. H_2SO_4 acid gives severe skin burns because it removes water and elements of water from skin tissue.



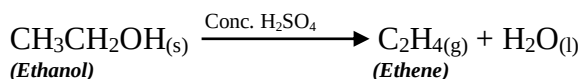
- Should the acid spill on skin, it is washed immediately with plenty of **water** followed with a solution of **sodium hydrogen carbonate**.
- Holes appear where the acid spills on clothes for same reason.

(d). Action on alcohols (alkanols)

- Conc. sulphuric (VI) acid dehydrates alcohols to corresponding alkenes.

Example: dehydration of ethanol to ethene

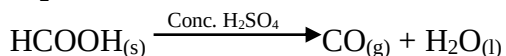
Equation:



(e). Action on methanoic acid.

- Conc. sulphuric (VI) acid dehydrates methanoic acid to form CO.

Equation:



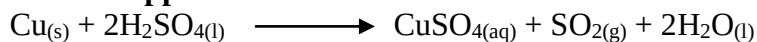
6. Further reactions of conc. H₂SO₄ as an oxidizing agent.

- Hot concentrated Sulphuric acts as an oxidizing agent in which cases it is reduced to sulphur (IV) oxide and water.

Examples:

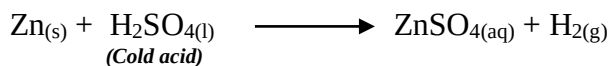
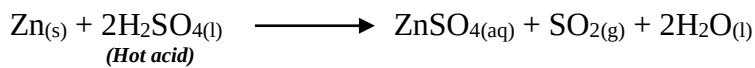
(a). Reaction with metals.

➤ Copper

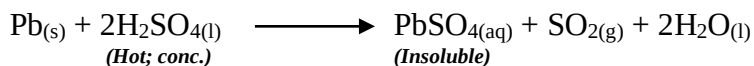


Note: the copper (II) sulphate formed is white since the conc. H₂SO₄ further dehydrates the hydrated CuSO₄.

➤ Zinc



➤ Lead



Note:

- **Dilute** sulphuric (VI) acid doesn't have any action on copper.

Reason:

- Copper is below hydrogen in reactivity series hence cannot displace it from the acid.
- This acid (H₂SO₄) has very little effects on lead, and usually the amount of SO₂ liberated is very little.

Reason:

- Formation of an **insoluble lead sulphate** layer that forms a protective coating on the metal stopping further reaction.



(b). Reaction with non-metals.

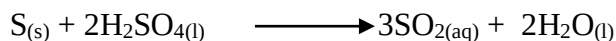
- Concentrated sulphuric acid **oxidizes** non-metals such as sulphur and carbon to their **respective oxides**.

Equations:

➤ With carbon



➤ With sulphur



8. It is a **less volatile acid**; and displaces more volatile acids (refer to lab preparation of HNO_3)

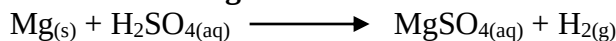
Reactions of dilute sulphuric acid

1. Reaction with metals

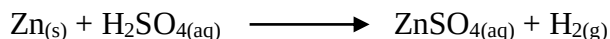
- It reacts with metals **above hydrogen** in the reactivity series to produce a **salt** and **hydrogen**.
- With potassium and sodium, reaction is **violent**.

Equations:

➤ With magnesium:



➤ With zinc:



Note:

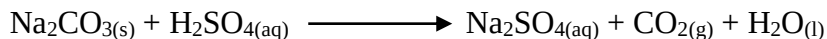
- Copper is below hydrogen in reactivity series hence can't displace hydrogen from dilute sulphuric (VI) acid.

2. Reaction with carbonates and hydrogen carbonates

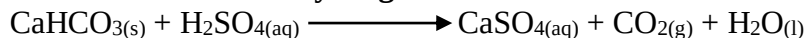
- Dilute $\text{H}_2\text{SO}_{4(aq)}$ reacts with carbonates and hydrogen carbonates to produce a salt, carbon (IV) oxide and water.

Equations

➤ With sodium carbonate:



➤ With calcium hydrogen carbonate:



Note:

- Reaction with lead carbonate however stops soon after the reaction.

Reason:

- Formation of an insoluble coating of the lead (II) sulphate on the lead (II) carbonate which prevents further contact between acid and carbonate.
- The same logic applies for calcium carbonate.

- Reacts to form salt and water.
- However, those metal oxides whose sulphates are insoluble react only for a while.
- Thus reaction between dilute sulphuric (VI) acid and lead (II) oxide stops almost immediately.
- This is due to formation of an insoluble layer of lead (II) sulphate which effectively prevents further contact between acid and oxide.

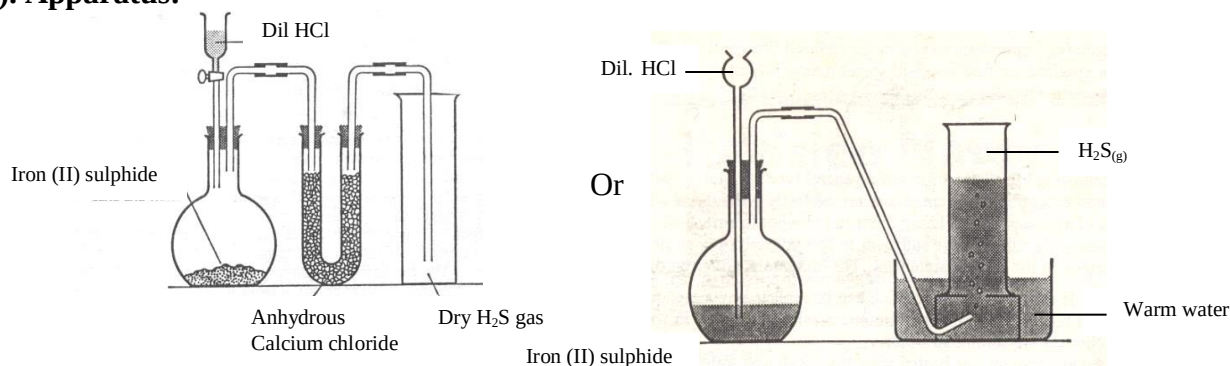
1. Manufacture of fertilizers.
2. Processing of metal ores.
3. Manufacture of detergents.
4. Manufacture of plastics.
5. Manufacture of dyes and paints.
6. Manufacture of lead and accumulators.
7. Manufacture of polymers.
8. Manufacture of petroleum (petroleum refinery).
9. Drying agent in industrial processes.

Hydrogen sulphide gas

- It is a **colourless gas** with a characteristic “**rotten egg**” smell; and is usually given out by rotting cabbage and eggs.

Laboratory preparation

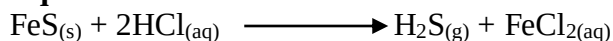
(i). Apparatus:



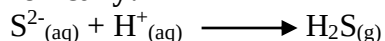
(ii). Procedure:

- Dilute hydrochloric acid is poured into Iron (II) sulphide in a round-bottomed flask.
- Resultant gas is passed through U-tube with anhydrous calcium chloride to dry the gas.
- This can also be done with **phosphorous (V) oxide**.

Equation:



Ionic:



(iii). Collection of gas

- When dry, the gas is collected by **downward delivery** because it is **denser** than air.
- When wet is collected over **warm water** because it is more soluble in cold water.

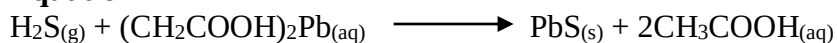
➤ Hydrogen sulphide test.

- When a strip of filter paper soaked in aqueous **lead (II) ethanoate** is put in hydrogen sulphide, the paper turns **black or dark brown**.

Reason:

- Due to the formation of lead (II) sulphide which is black.

Equation



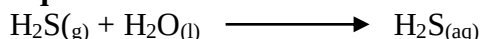
Properties of hydrogen sulphide gas



Physical properties

1. Colourless and very poisonous gas (similar to hydrogen cyanide)
2. Has a **repulsive** smell (similar to that of rotten eggs or decaying cabbages)
3. **Soluble** in water giving a weak acid (only slightly ionized)

Equation:

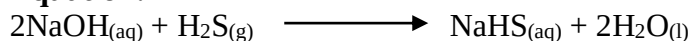


Then:



- The acid is **dibasic** hence forms hydrogen sulphides.

Equation:



Note:

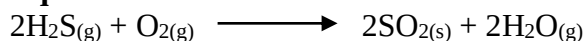
- Potassium hydroxide reacts similarly like sodium hydroxide.

Chemical properties

4. Combustion

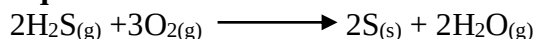
- Burns in a **blue flame** in a limited supply of oxygen (air) forming a **yellow deposit** of sulphur and steam.

Equation:



- In plentiful supply (excess) of Oxygen (air) it burns with a blue flame forming SO_2 and steam.

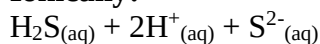
Equation:



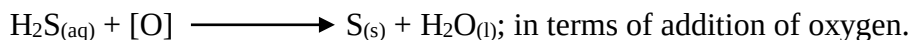
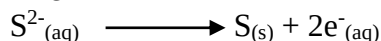
5. It is a reducing agent

- It supplies electrons which are accepted by the oxidizing agent and forms sulphur.

Ionic:



Then

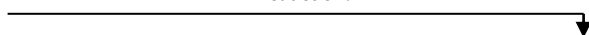


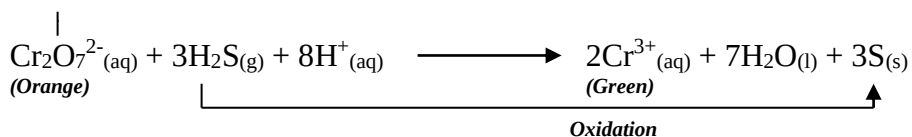
Examples

(i). With acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution (potassium dichromate VI)

Equation:

Reduction:

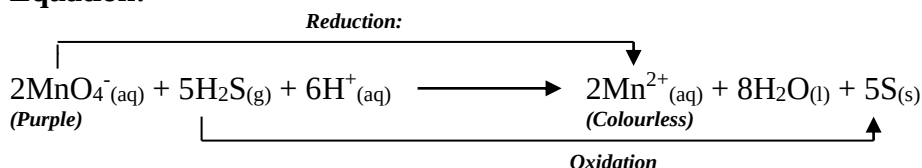




Observation: The orange solution turns green and H_2S oxidized to **yellow sulphur**.

(ii). Potassium manganate (VII) (KMnO_4)

Equation:

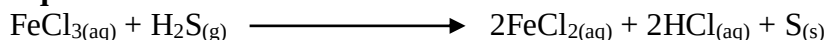


Observation:

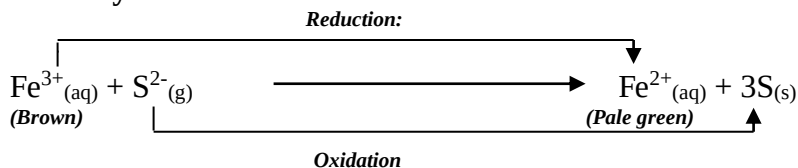
- The Purple solution turns colourless
- Manganate (VII) ions are reduced to manganate (II) ions; H_2S oxidized to **yellow sulphur**.

(iii). Action on Iron (III) chloride ions

Equation:



Ionic:



Observation:

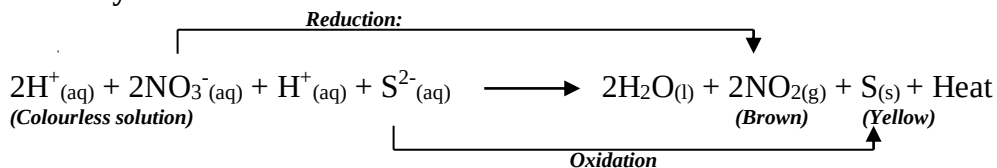
- The brown solution turns pale green;
- The $\text{Fe}^{3+}(\text{aq})$ are reduced to $\text{Fe}^{2+}(\text{aq})$; while the $\text{S}^{2-}(\text{aq})$ are oxidized to yellow sulphur.

(iv). Action with Conc. HNO_3

Equation:



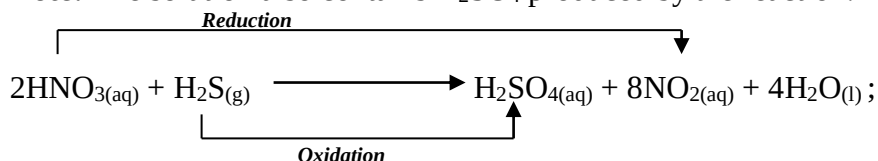
Ionic:



Observation:

- Evolution of brown fumes; and deposits of a yellow solid;
- $\text{HNO}_3(\text{aq})$ is reduced to **brown $\text{NO}_2(\text{g})$** ; while $\text{S}^{2-}(\text{aq})$ are oxidized to **yellow sulphur**;

Note: The solution also contains H_2SO_4 produced by the reaction:

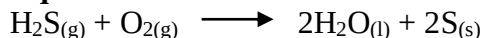




(v). Action of air on H₂S

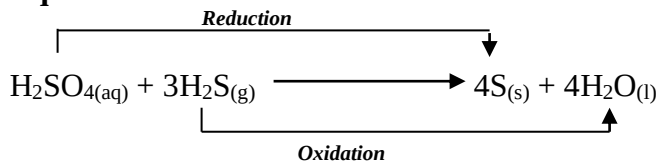
- The gas is dissolved in distilled water in a beaker and exposed to air; after a few days, a **white disposal** is formed.

Equation:



(vi). Action with concentrated sulphuric (VI) acid.

Equation

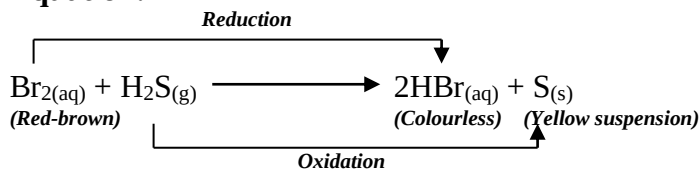


(vii). Action with halogen elements

➤ **Red-brown bromine water**

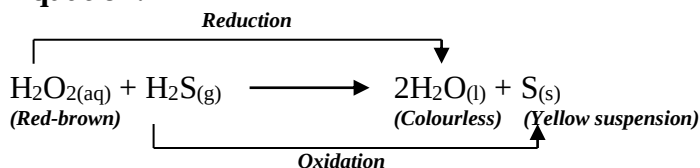
- Red-brown bromine water is **reduced** forming colourless hydrogen bromide (Hydrobromic acid) and **yellow** deposits (suspension) of sulphur.

Equation:



(viii). Action with hydrogen peroxide.

Equation:





Preparation of metallic sulphides

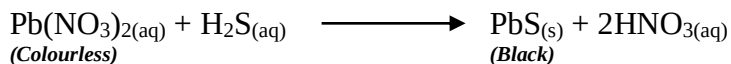
- Hydrogen sulphide reacts with metal ions in solution to form precipitates of metal sulphides; majority of which are black in colour.

(i). Procedure

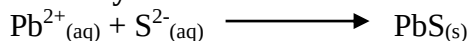
- The gas is bubbled through solutions of the following salts: $\text{Pb}(\text{NO}_3)_2$, CuSO_4 , FeSO_4 etc.

(ii). Observations and equations

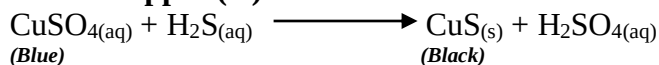
➤ Lead ions:



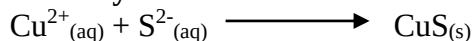
Ionicallly:



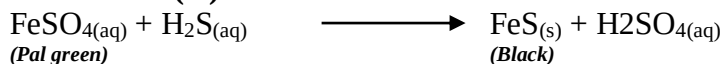
➤ Copper (II) ions:



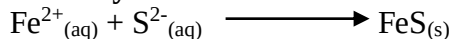
Ionicallly:



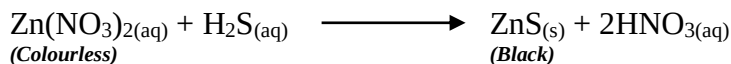
➤ Iron (II) ions:



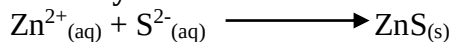
Ionicallly:



➤ Zinc ions:



Ionicallly:



Note:

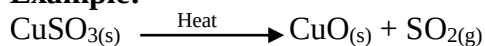
- Most metal sulphides are **insoluble** in water except those of sodium, potassium and ammonium.

- Are compounds of the sulphite radical (SO_3^{2-}) and a metallic or ammonium cation

Effects of heat

- They decompose on heating, forming SO_2 ;

Example:



Test for sulphites

(i). Procedure

- To 2cm³ of the test solution, add 2 cm³ of BaCl₂ or Ba (NO₃)₂; i.e. addition of barium ions.

- To the mixture add 2 cm³ of dilute HCl or HNO₃.

(ii). Observation

- A **white precipitate** (BaSO_3) is formed which **dissolves on addition of acid**.

- Production of a colourless gas that turns filter paper soaked in acidified **orange** potassium dichromate (VI) to **green**.

(iii). Explanations

- Only BaSO_3 ; BaCO_3 and BaSO_4 form white precipitates;

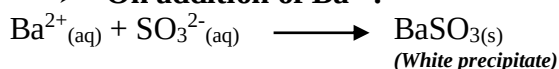
- The precipitates of BaSO_3 and BaCO_3 dissolve on addition of dilute acids; unlike BaSO_4 ;

- BaSO_3 produces $\text{SO}_{2(g)}$ as it dissolves on addition of a dilute acid; SO_2 turns orange acidified potassium dichromate (VI) to green;

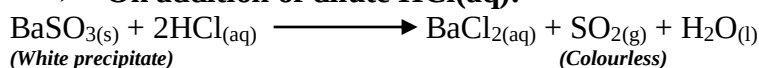
- BaCO_3 of the other hand dissolves in dilute acids producing CO_2 ; which has no effect on $\text{K}_2\text{Cr}_2\text{O}_7$; but forms a white precipitate in lime water;

Equations:

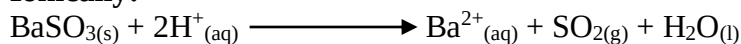
➤ **On addition of Ba^{2+} :**



➤ **On addition of dilute HCl(aq):**



Ioniclly:





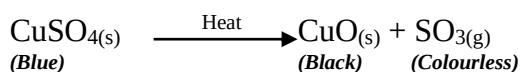
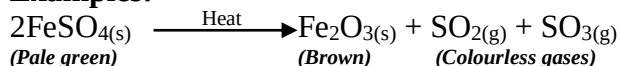
Sulphates

- Are compounds of the sulphate radical (SO_4^{2-}) and a metallic or ammonium cation.

Effects of heat.

- Decompose on heating and liberate SO_2 and SO_3 or SO_3 alone;
 - However quite a number of sulphates do not decompose on heating; and thus require **very strong heating** in order to decompose.

Examples:



Action of acids

Test for sulphates

- To about 2 cm^3 of the test solution, 2 cm^3 of BaCl_2 or $\text{Ba}(\text{NO}_3)_2$ solution is added.
 - To the mixture, 2 cm^3 of dilute HCl or HNO_3 is added.

Observation

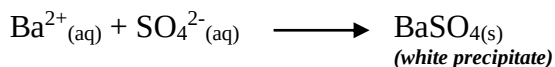
- A white precipitate is formed when $\text{Ba}(\text{NO}_3)_2$ is added; which is insoluble in excess acid.

Explanations.

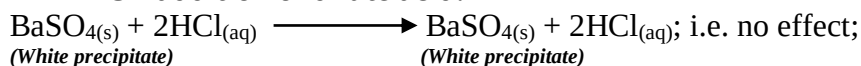
- Only BaSO_3 ; BaCO_3 and BaSO_4 form white precipitates;
 - The precipitates of BaSO_3 and BaCO_3 dissolve on addition of dilute acids; unlike BaSO_4 ;
 - Thus the white precipitate insoluble in dilute HCl or HNO_3 could only be a **sulphate**; in this case barium sulphate.

Equations:

➤ On addition of Ba^{2+} :



➤ On addition of dilute acid:



**Pollution by sulphur compounds.**

- Main pollutants are **sulphur (IV) Oxide** and **hydrogen sulphide**.

(a). Sulphur (IV) oxide.

- SO_2 is emitted when sulphur-containing fuels are burnt; during extraction of metals like copper and in manufacture of sulphuric (VI) acid.

- SO_2 is oxidized to SO_3 ;

- SO_3 reacts with water in atmosphere to form sulphuric (VI) acid which comes down as acid rain or acid fog.

Acid rain (fog) has environmental effects:

- Leaching of minerals in soil;
- Erosion of stone work on buildings;
- Corrosion of metallic structures;
- Irritation of respiratory systems thus worsening respiratory illnesses;
- Death of plants as a result of defoliation (falling of leaves);
- Destruction of aquatic life in acidified lakes;
- Stunted plant growth due to chlorosis;

(b). H_2S is very poisonous.



UNIT 5: CHLORINE AND ITS COMPOUNDS.

Unit Checklist:

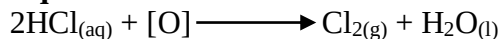
1. About chlorine.
2. Preparation of chlorine.
3. Properties of chlorine.
 - Colour and smell
 - Solubility in water
 - Action on litmus paper
 - Bleaching action
 - Action on hot metals
 - Reaction with non-metals
 - Oxidation reactions
 - Reaction with alkalis
 - Effect of sunlight on chlorine water.
4. Industrial manufacture of chlorine (The mercury cathode cell)
5. Uses of chlorine and its compounds
6. Hydrogen chloride gas
 - Preparation
 - Properties
7. Test for chlorides.
8. Hydrochloric acid
 - Large scale manufacture
 - Uses of hydrochloric acid
9. Environmental pollution of chlorine and its compounds

**Introduction:**

- Chlorine is a molecular non-metallic element made up of diatomic molecules.
- Its electron arrangement is 2.8.7 and it belongs to the halogen family.

Preparation of chlorine.

Note: It is usually prepared by oxidation of concentrated hydrochloric acid by removal of hydrogen.

Equation:

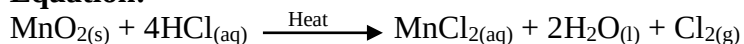
- The [O] is from a substance containing oxygen.

(a). Preparation of chlorine from MnO_2 and HCl .**(i). Apparatus:****(ii). Conditions:**

- Heating;
- Presence of an oxidizing agent; in this case it is manganese (IV) oxide.

(iii). Procedure:

- Hydrochloric acid is reacted with manganese (IV) oxide (dropwise);

Equation:**(iv). Explanation:**

- Manganese (IV) oxide oxidizes hydrochloric acid by removing hydrogen resulting into chlorine.
- The manganese (IV) oxide is reduced to water and manganese chloride.
- The resultant chlorine gas is passed through a bottle containing water.

Reason:

- To remove hydrogen chloride fumes (gas) which is very soluble in water.
- Next it is passed through concentrated sulphuric acid or anhydrous calcium chloride; to dry the gas.

(v). Collection:

- (a). Wet chlorine is collected over brine (saturated sodium chloride solution) or hot water.

Reason:

- It does not dissolve in brine and is less soluble in water

- (b). Dry chlorine is collected by downward delivery (upward displacement of air)

Reason:

- It is denser than air (2.5 times).

Note:

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- Chlorine may also be dried by adding calcium chloride to the jar of chlorine.

(c). The first bottle must contain water and the second concentrated sulphuric acid.

Reason:

- If the gas is first passed through concentrated sulphuric acid in the first bottle then to the water; it will be made wet again.

Properties of chlorine gas.

1. Colour and smell.

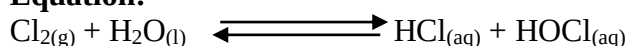
Caution: Chlorine is **very poisonous**.

- It is a green-yellow gas with an irritating pungent smell that attacks the nose and the lungs.
- It is 2.5 times denser than air, hence can be collected by downward delivery.

2. Solubility in water.

- It is fairly soluble in water forming green-yellow chlorine water.

Equation:



- Chlorine water is composed of two acids; chloric (I) acid (hypochlorous acid) and hydrochloric acid.

3. Action on litmus paper.

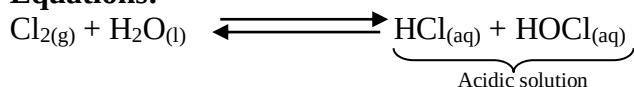
- Moist chlorine turns litmus paper red then bleaches it.
- Dry chlorine turns damp blue litmus paper red then bleaches it.
- Moist chlorine bleaches red litmus paper; dry chlorine bleaches damp red litmus paper.
- Dry chlorine has no effect on dry litmus paper.

Reasons:

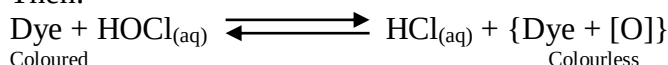
(i). In presence of moisture chlorine forms chlorine water which is acidic and hence turns blue litmus paper red.

(ii). Hypochlorous acid in the chlorine water is an oxidizing agent; thus adds oxygen (oxidizes) to the colour of most dyes; hence bleaching it.

Equations:



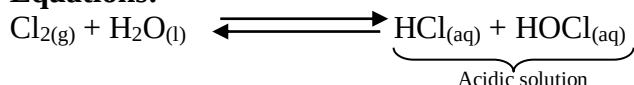
Then:



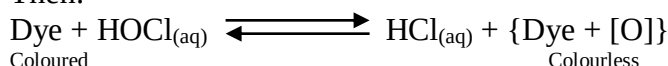
4. Bleaching action.

- Moist chlorine bleaches dyes but not printers ink which is made of **carbon**.
- The colour change is due to oxidation by hypochlorous acid.

Equations:



Then:



5. Action on a burning splint.

- The gas **put out** a glowing splint. It does not burn.



6. Action on hot metals.

(a). Preparation of iron (III) chloride.

(i). Apparatus.

(ii). Precaution.

- Experiment should be done in a fume cupboard or in the open.

Reason:

- Chlorine gas is poisonous and will thus be harmful to the human body.

(iii). Procedure:

- Dry chlorine gas is passed over iron wool as per the diagram.

(iv). Conditions.

➤ Chlorine gas has to be dry (done by the anhydrous calcium chloride in the U-tube)

Reason:

To prevent **hydration** hence **oxidation** of iron (which will then form $\text{Fe}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$) hence preventing reaction between iron and chlorine.

➤ Iron metal must be **hot**; and this is done by heating.

Reason:

To provide **activation energy** i.e. the minimum kinetic energy which the reactants must have to form products.

➤ Anhydrous calcium chloride.

- In the U-tube; to **dry** the chlorine gas.

- In the thistle funnel; to prevent **atmospheric water vapour** (moisture) from getting into the apparatus and hence reacting with iron (III) chloride.

(v). Observations:

- Iron metal glows red-hot.

- Red brown fumes ($\text{FeCl}_{3(g)}$) are formed in the combustion tube.

- A black solid ($\text{FeCl}_{3(s)}$) is collected in the flask.

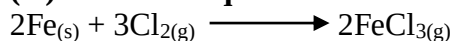
Note:

- Iron (III) chloride cannot be easily collected in the combustion tube.

Reason:

- It sublimes when heated and hence the hotter combustion tube causes it to sublime and its vapour is collected on the cooler parts of the flask.

(vi). Reaction equation.



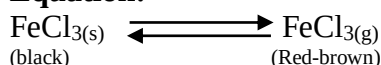
(vii). Conclusion.

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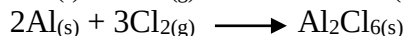
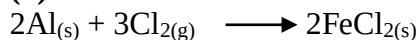


- Iron (III) chloride sublimes on heating; the black solid changes to red-brown fumes on heating.

Equation:



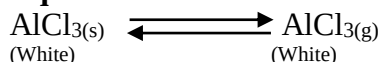
(b). Aluminium chloride.



Note:

- Aluminium chloride also **sublimes** on heating.

Equation:



(c). Reaction with burning magnesium.

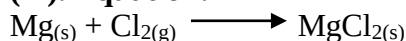
(i). Procedure:

- Burning magnesium is lowered into a gas jar of chlorine gas.

(ii). Observations:

- The magnesium continues to **burn** with a **bright blinding flame**;
- Formation of **white fumes** (MgCl_2); which cools into a white powder.

(iii). Equation:



- Generally chlorine reacts with most metals when hot to form corresponding chlorides.

Note:

Where a metal forms two chlorides when it reacts with chlorine, the higher chloride is usually formed.

Reason:

The higher chloride is **stable**. This explains why reactions of chlorine with iron result into iron (III) chloride and not iron (II) chloride.

7. Reaction with non-metals.

- It reacts with hot metals; forming covalent molecular compounds.

(a). Reaction with phosphorus.

(i). Procedure:

- A piece of warm phosphorus is lowered into a gas jar of chlorine.

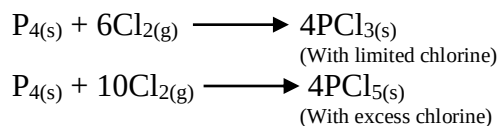
(ii). Observations:

- Phosphorus begins to **smoulder** and then ignites spontaneously.
- Evolution of **white fumes** (PbCl_3 and PCl_5)

(iv). Explanation.

- Chlorine reacts with warm dry phosphorus to form white fumes of phosphorus (III) and (V) chlorides.

Equations:



(b). Reaction with hydrogen.

(i). Conditions:

- Heating or **presence of light**; since chlorine and hydrogen do not react with each other at room temperature.

(ii). Precaution:

- The experiment is performed in a **fume chamber** (cupboard); since the reaction is **explosive**;

(iii). Procedure:

- Chlorine gas is mixed with hydrogen gas and the mixture heated or exposed to direct light; then aqueous ammonia brought near the mouth of the jar.

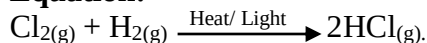
(iv). Observations:

- **White fumes** at the mouth of the jar.

(v). Explanations:

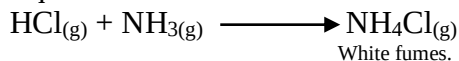
- Chlorine reacts explosively with hydrogen to form hydrogen chloride gas.

Equation:



- The hydrogen chloride gas diffuses upwards and reacts with ammonia at the mouth of the test tube to form white fumes of ammonium chloride; NH_4Cl .

Equation:



8. Chlorine as an oxidizing agent.

- Chlorine is a strong **oxidizing agent** and oxidizes many ions, by readily accepting electrons.
- During the process, chlorine itself undergoes reduction.

(a). Reaction with hydrogen sulphide gas.

(i). Procedure:

- A gas jar full of chlorine gas is inverted into another containing hydrogen sulphide gas.

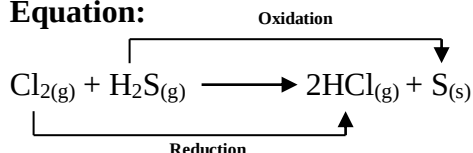
(ii). Apparatus:

**(iii). Observations:**

- **Yellow** deposits (of sulphur)
- **Misty** fumes (hydrogen chloride gas)

(iv). Explanations:

- Chlorine oxidizes hydrogen sulphide gas to sulphur solid, while itself is reduced to hydrogen chloride gas.

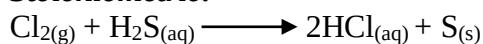
Equation:**(v). Conditions:**

- At least one of the gases must be **moist**; they do not react with each other in absence of moisture.

Note:

- In absence of moisture both gases are still in molecular form and hence cannot react; water facilitates their ionization hence ability to react.

- If aqueous hydrogen sulphide is used, then sulphur forms as a **yellow suspension** on the acidic solution.

Equations:**Stoichiometric:****Ionic:****(b). Reaction with sodium sulphite.****Procedure:**

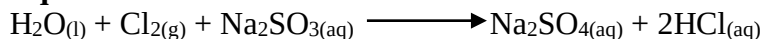
- Chlorine gas is bubbled through sodium sulphate in a beaker.
- Resulting solution is then divided into two portions.
- To the first portion, drops of dilute nitric acid are added followed by few drops of barium nitrate solution.
- To the second portion, few drops of lead (II) nitrate are added and the mixture warmed then cooled.

(ii). Observations:

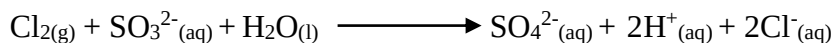
1st portion: White precipitate formed indicating presence of SO_4^{2-} ;

Explanations:

- The white precipitate indicate presence of SO_4^{2-} ; the precipitate is barium sulphate $\text{Ba}(\text{SO}_4)_2$;
- Chlorine oxidizes SO_3^{2-} in Na_2SO_3 to SO_4^{2-} while itself is reduced to chloride ions;

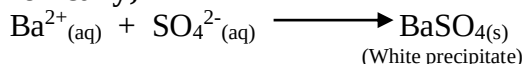
Equations:**Ionic:**

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- On adding barium nitrate (**Ba(NO₃)₂**); the **Ba²⁺** ions react with the **SO₄²⁻** to form insoluble **BaSO₄**; the white precipitate.

Ioniclly;



Note:

- The solution is first acidified (with HNO₃) before addition of **Ba(NO₃)₂** to prevent **precipitation** of **BaSO₃(s)** and **BaCO₃(s)**.

2nd portion:

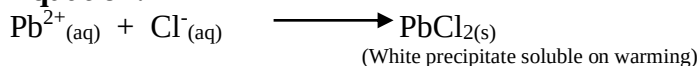
Observation:

- Formation of a white precipitate on addition of **Pb(NO₃)₂** solution.
- On warming the white precipitate dissolves then recrystalizes back on cooling.

Explanations:

- The white precipitate shows presence of either Cl⁻; SO₃²⁻ or SO₄²⁻
- However the fact that it dissolves on warming confirms the presence of Cl⁻(aq) and not SO₃²⁻(aq) and SO₃²⁻(aq)

Equation:



(c). Reaction with ammonia.

(i). Procedure:

Chlorine gas is bubbled through aqueous ammonia.

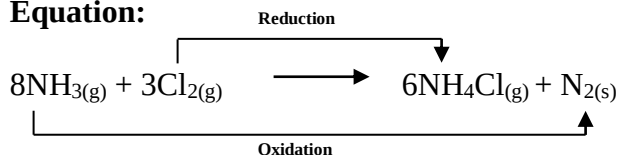
(ii). Observations:

- Evolution of **white fumes**.

(iii). Explanation.

- Chlorine gas oxidizes ammonia to nitrogen, while is itself reduced to white fumes of ammonium chloride.

Equation:



(d). Displacement reactions with other halogens.

(i). Procedure:

- Chlorine is bubbled through aqueous solutions of fluoride, bromide and iodide ions contained in separate test tubes.

(ii). Observations and explanations:

➤ **With fluoride ions.**

- No observable change or no reaction; because chlorine is a **weaker oxidizing agent** than fluorine.



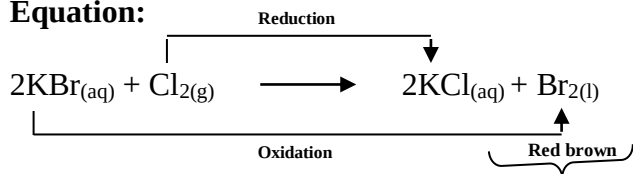
➤ **With bromide ions:**

- If potassium bromide was used, the colourless solution turns red-brown.

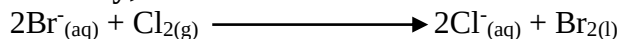
Reason:

- Chlorine has a higher tendency to **gain electrons** than bromine.
- It readily oxidizes bromide ions (in KBr) to form potassium chloride and bromine which immediately dissolves to make the solution **red-brown**.

Equation:



Ionically;



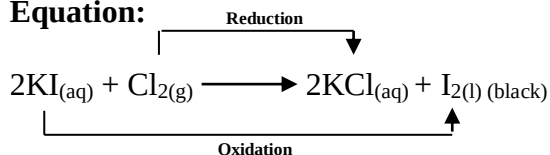
With iodide ions.

- Using potassium iodide the colourless solution would turn black.

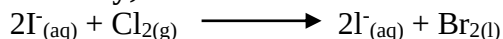
Reason:

- Chlorine has a higher tendency to gain electrons than iodine.
- It readily oxidizes the I^{-} (in KI) to form iodine and potassium chloride.
- Iodine solid in the resulting solution makes it black.

Equation:



Ionically;



9. Reaction with alkalis.

(a). Reaction with sodium hydroxide solution.

(i). Procedure:

- Bubble chlorine slowly through cold dilute sodium hydroxide solution.
- Dip litmus paper.

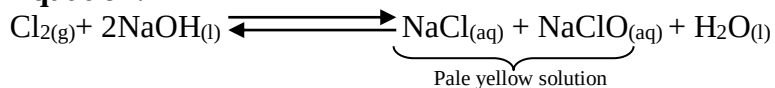
(ii). Observation:

- Litmus paper is bleached; the product has the colour and smell of chlorine.

(iii). Explanation:

- Chlorine dissolves in sodium hydroxide to form a pale yellow solution of sodium chlorate (I) or sodium hypochlorite (NaClO);
- The sodium chlorate (I) bleaches dyes by oxidation.

Equation:

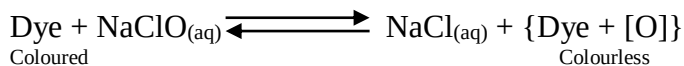


Bleaching action of NaClO :

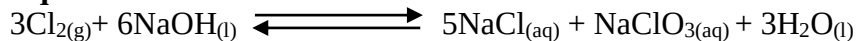
- The NaClO donates **oxygen** to the dye making it **colourless**; and thus it bleaches by **oxidation**.

Equation:

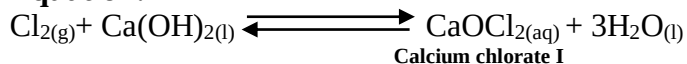
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**Note:**

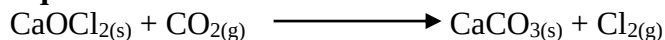
With hot concentrated sodium hydroxide, the chlorine forms **sodium chlorate (III); NaClO₃**.

Equation:**(b). Reaction with potassium hydroxide**

- Follows the trend of sodium.

(c). Reaction with slaked lime {Ca(OH)_{2(s)}}**Equation:****Note:**

Bleaching powder, CaOCl₂ always smells of strongly of chlorine because it reacts with carbon (IV) oxide present in the atmosphere to form chlorine.

Equation:**10. Effects of chlorine gas on:****(a). A burning candle.****(i). Procedure:**

- A burning candle is lowered into a gas jar of chlorine.

(ii). Observations:

- It burns with a small, **red** and **sooty flame**.

(iii). Explanations:

- Wax (in candles) consists of mainly hydrocarbons.
- The hydrogen of the hydrocarbon reacts with chlorine forming hydrogen chloride while leaving behind carbon.

(b). warm turpentine.**(i). Procedure:**

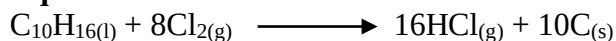
- A little turpentine is warmed in a dish and a filter paper soaked (dipped) in it.
- The filter paper is then dropped into a gas jar of chlorine.

(ii). Observation:

- There is a **red flash** accompanied by a **violent action** whilst a **black cloud** of solid particles form.

(iii). Conclusion:

- Black cloud of solid is carbon.
- Turpentine (a hydrocarbon) consists of hydrogen and carbon combined together.
- The chlorine combines with hydrogen and leaves the black carbon behind.

Equation:**11. Effects of sunlight on chlorine water.**

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**(i). Procedure:**

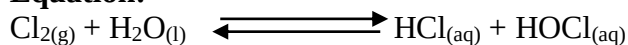
- Chlorine water is made by dissolving the gas in water.
- A long tube filled with chlorine water is inverted over a beaker containing water.
- It is then exposed to sunlight (bright light) as shown below.

(ii). Apparatus:**(iii). Observations:**

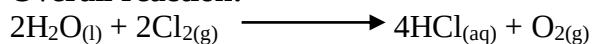
- After sometime a gas collects in the tube and on applying a glowing splint, the splint is rekindles showing that the gas collected is oxygen.

(iv). Explanation:

- Chlorine water has two components.

Equation:

- The HOCl being unstable will dissolve on exposure to sunlight, giving out oxygen.

Equation:**Overall reaction:****Industrial manufacture of chlorine (the mercury cathode cell)****The electrolysis of brine****(i). Apparatus.**

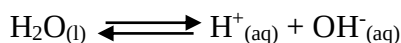
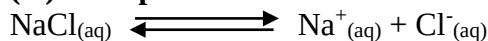
**(ii). Electrolyte.**

- **Brine**, concentrated sodium chloride solution, NaCl

(iii). Electrodes.

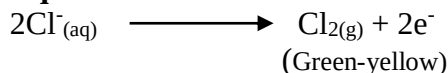
Anode: carbon (graphite)

Cathode: Flowing mercury;

(iv). Ions present:**(v). Reactions:****Anode:**

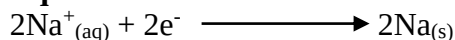
- Cl^- and OH^- migrate to the anode.

- Because of **high concentration** of $\text{Cl}^-_{(\text{aq})}$, they are discharged in preference to OH^- ions.

Equation:**Cathode:**

- $\text{H}^+_{(\text{aq})}$ and $\text{Na}^+_{(\text{aq})}$ migrate to the cathode.

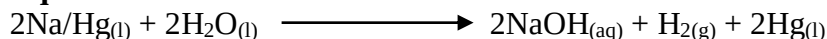
- Because the cathode is made of mercury, $\text{Na}^+_{(\text{aq})}$ is discharged in preference to $\text{H}^+_{(\text{aq})}$ ions;

Equation:**Note:**

- Sodium formed at the cathode dissolves in the flowing mercury cathode to form **sodium amalgam** (Na/Hg).

- Sodium amalgam is reacted with water to form sodium hydroxide and hydrogen.

- Mercury (in the sodium amalgam) remains unreacted.

Equation:

- The unreacted mercury is recycled.

(vi). Products:

- **Chlorine** gas at the anode.

- **Hydrogen** and **sodium hydroxide** at the cathode.

Uses of chlorine gas and its compounds.

1. Manufacture of **hydrochloric acid**.

2. Used in form of **bleaching powder** in textile and paper industries.



3. For **sterilization** of water for both domestic and industrial use and in swimming pools.
4. Used in **sewage treatment** e.g. NaOClO₃ solution used in latrines.
5. Manufacture of **plastics** (polyvinyl chloride; PVC)
6. Manufacture of **germicides**, pesticides and fungicides e.g. DDT and some CFCs.
7. CFCs are used to manufacture **aerosol propellants**.
8. Manufacture of **solvents** such as trichloromethane and some chlorofluorocarbons (CFCs).
9. CFCs are commonly **freons** are used as **refrigerants** in fridges and air condition units due to their low boiling points.
10. Manufacture of **chloroform**, an aesthetic.

Hydrogen chloride gas.

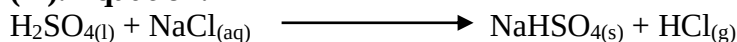
Laboratory preparation of hydrogen chloride gas.

(i). Apparatus:

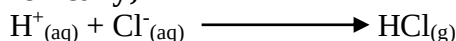
(ii). Procedure:

- Concentrated sulphuric acid is reacted with sodium chloride, and the mixture heated gently.
- Resultant gas is passed through conc. Sulphuric (VI) acid; to dry the gas.

(iii). Equation:



Ionic;



Note:

- The reaction can proceed in the cold, but on large scale HCl(g) is produced by the same reaction but the heating is continued to re hot.

Properties of hydrogen chloride gas.

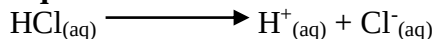
1. Colourless gas with a strong irritating pungent smell.
2. Slightly denser than air (1¼ times). This makes it possible to collect the gas by downward delivery.
3. Very soluble in water; and fumes strongly in moist air forming hydrochloric acid deposits.

Diagram:



- The aqueous solution is known as hydrochloric acid.
- It is almost completely ionized (a strong acid) in aqueous solution.

Equation:



- This solution has the usual acidic properties:

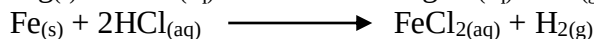
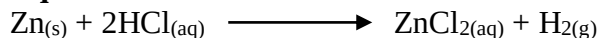
Examples:

- turns blue litmus red.
- Liberates hydrogen gas with certain metals e.g. zinc, Magnesium, iron etc.

Note:

Hydrochloric acid does not react with metals below hydrogen in the reactivity series.

Equations:



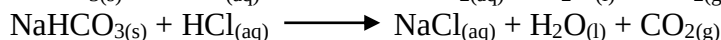
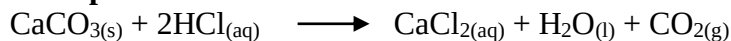
- Neutralizes bases to form salt and water.

Examples:



- Liberates carbon (IV) oxide from carbonates and hydrogen carbonates.

Examples:



Note:

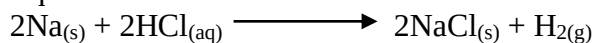
As the hydrogen chloride gas very soluble in water, the solution must be prepared using a funnel arrangement; to prevent sucking back and increase the surface area for the dissolution of the gas;

Diagram: dissolution of hydrogen chloride gas



4. Dry hydrogen chloride is NOT particularly reactive at ordinary temperatures, although very reactive metals burn in it to form the chloride and hydrogen gas.

Equation:



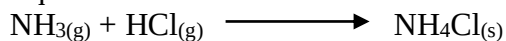
Metals above hydrogen in the reactivity series react with hydrogen chloride gas when heated.

Note:

If reacted with some metals it forms 2 chlorides e.g. iron where iron (II) and iron (III) chlorides exist.

5. Hydrogen chloride gas forms white fumes of ammonium chloride when reacted with ammonia gas;

Equation:



Note: This is the **chemical test** for hydrogen chloride gas.

6. Hydrogen chloride is decomposed by oxidizing agents, giving off chlorine.

Examples:

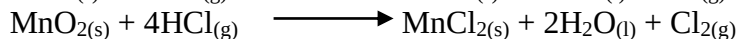
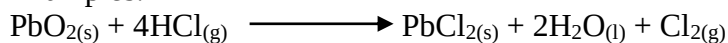


Diagram: reacting hydrogen chloride with an oxidizing agent.

Test for chlorides.

Test 1: Using silver ions:

Procedure:

- To the test solution, add silver ions from silver nitrate.
- Acidify with dilute nitric acid.

(ii). Observations and inference:

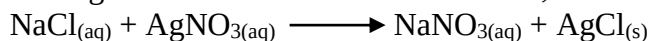
- Formation of a white precipitate shows presence of $\text{Cl}^-_{(aq)}$

(iii). Explanations:

- Only silver carbonate and silver chloride can be formed as white precipitates.
- Silver carbonate is soluble in dilute nitric acid but silver chloride is not.

Equations:

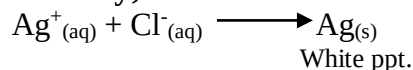
- Using Cl^- from NaCl as the test solution;





White ppt.

Ionicly;



Note:

- This precipitate dissolves in **excess ammonia**.
- The white precipitate of silver chloride turns **violet** when exposed to light.

Test 2: Using lead ions

(i) Procedure:

- To the test solution, add lead ions from lead (II) nitrate, then warm

(ii). Observations and inference:

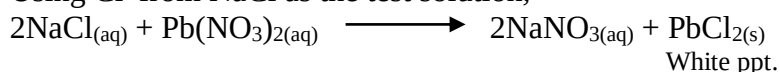
- Formation of a white precipitate that dissolves on warming shows presence of $\text{Cl}^-_{(\text{aq})}$

(iii). Explanations:

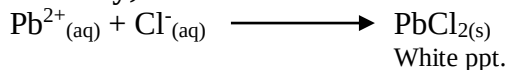
- Only lead carbonate, lead sulphate, lead sulphite and lead chloride can be formed as white precipitates.
- Only lead chloride dissolves on warming; unlike the rest which are insoluble even on warming.

Equations:

Using Cl^- from NaCl as the test solution;



Ionicly;



Hydrochloric acid.

Large scale manufacture of hydrochloric acid.

(i). Diagram:

(ii). Raw materials:

- **Hydrogen** obtained as a byproduct of petroleum industry; electrolysis of brine or from water by **Bosch process**;

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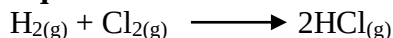


- **Chlorine** obtained from the electrolysis of brine or as fused calcium chloride.

(iii). Procedure:

- A small sample of hydrogen gas is allowed through a jet and burnt in excess chlorine gas.

Equation:



Precaution: A mixture of equal volumes of hydrogen and chlorine explodes when put in sunlight.

- The hydrogen chloride gas formed is dissolved in water over glass beads.
- The **glass beads increase the surface area** over which absorption takes place.
- Commercial hydrochloric acid is about 35% pure.
- Hydrochloric acid is transported in steel tanks lined inside with **rubber**.
- If the acid comes into contact with exposed parts of metal or with rust, it forms **iron (III) chloride** that makes the acid appear **yellow**.

Pollution in an industry manufacturing hydrochloric acid.

- (i). Chlorine is **poisonous**.
- (ii). Mixture of hydrogen and oxygen in air is **explosive** when ignited.

Uses of hydrochloric acid.

1. Sewage treatment.
2. Treatment of water (**chlorination**) at the waterworks.
3. Removing rust from metal e.g. descaling iron before it is galvanized or and other metals before they are electroplated.
4. Making dyes, drugs and photographic materials like **silver chloride** on photographic films.

Environmental pollution by chlorine and its compounds.

1. Chlorine may dissolve in rain and fall as **acid rain**, which has adverse effects on plants and animals, buildings and soil nutrients.
 2. CFCs are **non-biodegradable**. Over time, they diffuse into the atmosphere breaking down to free chlorine and fluorine atoms. These atoms deplete the **ozone layer**. Chlorine is thus one of the **greenhouse gases**.
 3. PVCs are **non-biodegradable**.
 4. DDT is a pesticide containing chlorine and has a long life span, affecting plants and animal life.
- Note: DDT is banned in Kenya; NEMA advises increased use of **pyrethroids** in mosquito control.